

Mapping Polarization in U.S. Party Platforms from 2008-2024 with Topic Modeling and Sentiment Analysis

Yeuk Wai Ting

December 14, 2025

Abstract

This study examines the evolution of thematic structure and rhetorical tone in U.S. party platforms from 2008 to 2024. Using party platform data from the Manifesto Project, a three-stage natural language processing pipeline that includes document embeddings, topic modeling, and sentiment analysis. The findings indicate that Democrats and Republicans increasingly emphasize distinct clusters of issues and frame shared domains, such as healthcare, immigration, and technology, in different ways. Additionally, the emotional tone associated with these topics has become more polarized. These patterns suggest that recent polarization encompasses more than ideological divergence; it also reflects a growing separation in how parties define political problems, construct governing agendas, and communicate them to the public. By mapping semantic and structural variation across five election cycles, this project aims to contribute to the understanding of how partisan conflict in the post-2008 political environment.

1 Introduction

Over the past several election cycles, American party platforms have begun to appear increasingly disconnected from one another. This is not only a matter of widening ideological distance. When we examine platforms from 2008 onward, a deeper pattern emerges. The major parties are not simply offering different policy solutions. They frequently describe different political problems and, in some cases, different political worlds. This contrast is visible even before any formal analysis. In 2024, for example, the Republican Party platform opens its immigration section with the headline: “Stop Illegal Immigration. Republicans will secure the Border, deport Illegal Aliens...” By contrast, Democratic leaders and platform summaries present immigration as part of a broader effort to “fix a broken immigration system” and uphold values of inclusion and fairness. These statements do not merely express disagreement about preferred policies. They reveal distinct definitions of what the issue itself represents, which raises a motivating question: when did the two parties cease to engage with a shared set of governing challenges.

Understanding this shift requires situating the platforms within the broader context of the post-2008 period. The financial crisis destabilized long-standing economic assumptions and public trust in institutions. It also generated new forms of economic and social precarity that altered expectations about the role of government. At the same time, geopolitical change and rapid technological innovation introduced policy concerns that earlier platforms addressed only minimally. These developments suggest that partisan evolution after 2008 may involve more than changes in ideological positioning. It may reflect a reorganization of how each party understands the central problems facing the United States. Yet existing research on American polarization tends to focus on ideological distance, affective divisions or elite sorting. Less attention has been paid to the possibility that the parties are diverging in their agenda structures and in the ways they organize and interpret issue domains.

This project begins from that observation. We examine whether the period from 2008 to 2024 reflects a form of agenda-level polarization in which the parties increasingly emphasize different clusters of issues, frame shared topics through different definitions and attach opposing emotional tones to policy debates. If this pattern is present, then contemporary polarization involves more than disagreement within a shared policy space. It reflects separation in how the parties conceptualize the challenges that national governance must address.

To investigate this possibility, we analyze U.S. party platforms using a three-stage natural language processing (NLP) pipeline applied to quasi-sentences from the Manifesto Project. Similarity analysis, topic modeling and transformer-based sentiment analysis allow us to trace both broad thematic changes and fine-grained variation in

the internal structure of specific policy areas. This approach enables us to assess not only which issues receive emphasis but how those issues are internally organized and rhetorically framed.

Guided by these motivations, we propose the following research questions:

- How have the thematic structures of Democratic and Republican platforms evolved from 2008 to 2024.
- Do the parties increasingly prioritize different issue clusters, indicating divergence in agenda construction.
- Within common issues, how do the parties differ in the internal composition and framing of their subtopics over time.
- How does the emotional tone associated with these topics evolve, and does it contribute to widening rhetorical distance.

These questions matter because party platforms remain one of the most formal statements of how political organizations interpret national challenges and communicate priorities to diverse audiences, including activists, donors, legislators and the electorate. Understanding how platform agendas shift over time provides insight into the foundations of contemporary political conflict and the conditions under which bipartisan problem-solving becomes more difficult. By tracing semantic, structural and affective change across sixteen years of platform writing, this project aims to determine whether the post-2008 era reflects not just sharper ideological disagreement, but an emerging separation in how the parties understand national issues.

2 Literature Review

2.1 The 2008 Critical Juncture

The period beginning in 2008 provides a useful context for understanding shifts in party platform agendas because it brought together several political, economic and generational developments that altered the environment in which parties communicated their priorities. These developments did not produce a sharp break in American politics, nor did they fundamentally transform party ideology overnight. However, they created new pressures and incentives that plausibly shaped how parties selected issues, framed policy domains and articulated their governing visions. Situating the analysis within this period therefore helps clarify why platform content may have evolved in the way we observe.

The first relevant development was the financial crisis. As Reinhart and Rogoff show, severe financial downturns are characterized by deep and prolonged asset-market collapses, with real housing prices declining an average of 35 percent over six years, substantial increases in unemployment and significant rises in public debt (Reinhart and Rogoff, 2009)[43]. Mian, Sufi and Trebbi provide empirical evidence that such crises heighten distributive conflicts and increase legislative gridlock, producing what they describe as a “political tug-of-war” between debtors and creditors. Creditors are often better organized and more effective at securing bailouts while broad-based debt relief is constrained, resulting in “anemic reform” (Mian, Sufi and Trebbi, 2012)[32]. These dynamics shape not only economic policymaking but also broader narratives around inequality, responsibility and trust in government. As Gottfried argues, the post-crisis environment provided fertile ground for a “populist oligarchy,” in which elites employed strategies of disruption and polarized rhetoric to manage democratic instability while deflecting anti-systemic anger (Gottfried, 2019)[13]. These shifts in tone, expectations and institutional confidence likely influenced the kinds of issues parties emphasized in their election platforms.

A second development concerned the geoeconomic environment. The financial crisis raised concerns about U.S. uni-polarity and highlighted limitations about the limitations of the Washington Consensus. Saray notes that while neoliberal reform increased profitability in the United States, it also exacerbated inequality and failed to deliver sustainable development in emerging markets, prompting interest in alternative models such as the “Beijing Consensus” (Saray, 2019).[44] This global shift had domestic implications. As Trubowitz and Harris argue, the core challenge to American international leadership during this period was not simply the rise of new rivals. More fundamentally, it was the weakening of domestic unity, which reduced the United States’ “usable power” by limiting the political foundations for bipartisan foreign-policy engagement (Trubowitz and Harris, 2019)[46]. This shift, in turn, deepened the divisions between the parties in their approaches to international affairs, undermining the possibility of a consistent and unified policy direction.

A third development involved generational change. Millennials entered the electorate during a period of economic insecurity and institutional uncertainty. Milkman describes this cohort as a “new political generation” shaped

by precarious labor markets and the erosion of the long-held belief that higher education would lead to stable, middle-class lives. The prevailing image of a 'graduate with no future' encapsulates the disillusionment and uncertainty that now define the prospects of many young people (Milkman, 2017).^[33] As digital natives, Millennials mobilized politically through social media, bypassing traditional gatekeepers and enabling rapid horizontal forms of organization. Occupy Wall Street and Black Lives Matter are great examples of such movements, which highlighted structural inequality, racial justice. These generational pressures may have affected parties unevenly. Democrats increasingly incorporated themes related to inequality, inclusion and social justice, while Republicans emphasized sovereignty, stability and cultural continuity. These divergent responses contributed to differences in how each party conceptualized emerging political and economic challenges.

Collectively, these developments reshaped the landscape in which political parties operate, forcing them to navigate evolving coalitions and respond to increasingly divided public demands. Although 2008 did not mark a formal break in the institutional order, the years that followed brought about circumstances that pressed parties to reconsider both the substance of their agendas and the language they used to present them. Against this backdrop, the central question emerges: to what extent does the transformation of party platforms between 2008 and 2024 signal a widening gap in how the major parties identify and rank the nation's most pressing policy challenges?

2.2 Party Platforms as Political Documents

Party platforms in the United States occupy a paradoxical position: they are both performative political documents and repositories of substantive party commitments. Historically, many observers dismissed platforms with the claim that candidates "run on" them rather than "stand on" them, implying that parties treat these documents as symbolic rather than binding. However, empirical research complicates this assumption. Studies find that presidents frequently attempt to fulfill platform pledges through legislative collaboration or executive action (Fishel 1985; Jacobs and Shapiro 2000)^{[10][24]}. Oliver and Marion (2008) similarly demonstrate that platform rhetoric, even when symbolic, offers meaningful cues that shape presidential priorities and communication.^[37] This contested status makes platforms especially valuable for understanding how parties articulate their ideological positions and interpret political challenges at particular historical moments.

Recognizing platforms as partly performative also highlights that they are crafted for multiple audiences, not just electorate. Platforms often speak just as much to interest groups, activists, donors, and legislative actors, many of whom exert substantial influence over the parties. Reinhardt and Victor (2009) show that platforms signal which constituencies the party views as electorally or organizationally critical.^[42] Harmel et al. (2018) further conceptualize platforms as addressing the "two faces" of parties: they communicate an external image aimed at voters while simultaneously strengthening an internal identity directed toward activists, members and rival factions.^[19] This dual purpose explains why platforms mix broad thematic rhetoric with detailed policy commitments.

Because platforms serve these multiple audiences, they inevitably reflect intra-party bargaining and factional pressures. Coalition-based theories of party organization (Aldrich 1995^[1]; Cohen et al. 2008^[8]) describe platforms as negotiated compromises aimed at unifying diverse groups under a shared policy vision. However, platforms do not always reflect coalition medians. Research by Oliver and Marion (2008)^[37] and Skocpol and Williamson (2012)^[45] shows that dominant factions, such as incumbents, primary victors, ideological movements, or well-organized activist blocs, can shape platform language disproportionately. One example is the dramatic shift in the Republican Party platform in 1980, which, driven by the increasing mobilization of the Christian right, reversed decades of policy to explicitly call for an antiabortion amendment and opposition to the Equal Rights Amendment (Lewis, 2019)^[26]. This episode illustrates how organized activism can imprint durable rhetorical themes onto formal party commitments. Thus, platform text is often shaped as much by internal negotiation and factional appeasement as by coherent programmatic planning.

Beyond accommodation of interests and coalition negotiation, platforms serve to build a consistent narrative and identity. They articulate the party's principles, frame national problems, and signal what the party views as politically important, even in cases where policy details remain vague or strategically incomplete. Oliver and Marion (2008) show how parties use symbolic rhetoric, such as "getting tough on crime," to signal alignment with public sentiment without elaborating detailed policy formulas.^[37] Harmel et al. (2018) argue that changes in rhetorical emphasis can indicate changes in the public image of the party, while changes in substantive position-taking reveal internal debates over identity.^[19] Meanwhile, Lo, Proksch, and Slapin (2016) show that manifestos encode structured ideological profiles, reinforcing that platform language is sufficiently systematic to allow meaningful

quantitative analysis.[27]

These theoretical insights underscore the value of platform texts as a foundation for analyzing party ideology and its evolution. While political parties employ a variety of communication channels—campaign speeches, debates, press releases, and, increasingly, social media—these outlets are typically reactive, tailored to individual candidates, or shaped by the exigencies of the moment. Such communications tend to reflect immediate electoral imperatives rather than the deeper institutional commitments of the party. In contrast, platforms are the product of collective negotiation among activists, interest groups, officeholders, and party elites. They formalize the party’s position after extensive negotiations, balancing symbolic gestures with substantive policy commitments, and setting forth priorities to emphasis on through an electoral cycle. Platforms thus offer a more stable and coherent signal of a party’s ideological direction than the fragmented and transient messages found elsewhere. For scholars and observers alike, this distinction is critical: only by focusing on these foundational documents can one hope to trace the underlying trajectory of party ideology with any degree of confidence.

2.3 Applying NLP Techniques to Platform Texts

Analyzing party platforms requires methodological tools that can detect changes in both policy emphasis and rhetorical framing across time. The challenge is that platforms are neither purely programmatic nor purely symbolic; they sit somewhere in between. This means that no single computational method can fully capture their ideological content. Recent advances in natural language processing offer new capabilities, but they also introduce assumptions and limitations that must be addressed directly. The growing methodological literature reflects precisely this tension: researchers seek tools that can scale to large corpora while still preserving the interpretive rigor expected in political science.

Contextual embeddings, particularly those produced by Sentence-BERT, represent one of the most significant methodological shifts in recent years. Their key advantage is that they encode words in context rather than as isolated units. This allows embeddings to capture semantic relationships that older bag-of-words approaches ignore. However, these models also come with risks: embedding spaces are shaped by the pretraining data and may inadvertently import biases or ideological priors. Scholars have nonetheless demonstrated that, when carefully validated, embeddings can recover ideological proximity that aligns closely with expert-coded benchmarks (Maurer et al. 2022[28]; Ceron et al. 2022[5]). Their value lies not in replacing interpretation, but in making patterns of rhetorical convergence and divergence visible at scale. The methodological consensus is not that embeddings solve the problem of ideological measurement, but that they expand the set of plausible tools when paired with appropriate checks.

A similar dynamic appears in the evolution of topic modeling. Traditional LDA-based models have been widely used because they provide a transparent generative framework. Yet political texts routinely violate LDA’s core assumptions: word independence, stable vocabularies, and topic separability. BERTopic improves coherence by clustering in embedding space, but it introduces new vulnerabilities: topic boundaries depend on clustering hyperparameters, and the interpretability of c-TF-IDF keywords varies across corpora. Recent studies nonetheless show that BERTopic produces more stable and substantively meaningful topics when applied to political texts, especially those that mix symbolic and policy language (Hellwig et al. 2024)[21]. Its contribution is not that it is inherently superior to LDA, but that it is better suited for capturing evolving, intersectional issue bundles that characterize post-2008 platform writing. Scholars increasingly treat BERTopic as one piece of a larger toolkit—useful but not definitive.

A growing strand of methodological research also argues that topic models alone are insufficient because they cannot distinguish between neutral discussion of an issue and its evaluative framing. This concern is well-documented in political communication research, which shows that parties may address the same issues but deploy them in rhetorically incompatible ways. Transformer-based sentiment classifiers respond to this gap by capturing valence and emotional tone, though they too require caution: sentiment models trained outside political contexts often misread partisan idioms or sarcasm. When fine-tuned appropriately, however, they outperform lexicon methods and reveal differences in evaluative framing that topic clusters cannot detect (Hellwig et al. 2023[20]; Orellana & Bisgin 2023[38]). Importantly, the methodological literature does not treat sentiment as a substitute for ideology, but as an independent dimension that interacts with issue emphasis. This distinction becomes particularly salient in high-polarization environments where emotional appeals structure partisan identity.

Across these studies, one pattern is clear: contemporary political texts require multidimensional analysis. Embeddings reveal semantic structure; contextual topic models identify how parties distribute attention; sentiment tools

capture evaluative tone. Each method has strengths and vulnerabilities, and none produces interpretable results without theoretical guidance and validation. Grimmer and Stewart (2013) warn against assuming that automated methods “speak for themselves,” emphasizing that models should be treated as measurement strategies embedded within substantive theory.[14] The field’s movement toward combined approaches reflects this caution. Rather than claiming that new models supersede earlier ones, scholars argue that they allow us to interrogate ideological change with greater granularity—provided we remain explicit about what each method can and cannot measure.

For examining U.S. party platforms between 2008 and 2024, the methodological literature therefore supports an integrated framework. The goal is not to rely on computational models as definitive arbiters of ideology but to use them as systematic ways of detecting patterns that require further interpretation. This layered approach aligns with both the opportunities and the constraints emphasized in recent text-as-data scholarship: the need to balance scale with meaning, complexity with clarity, and methodological innovation with theoretical discipline.

3 Data and Methods

This section introduces the party platforms data used in this study and the three-stage natural language processing (NLP) pipeline used to examine these data. Party platforms are long documents covering multiple issues and facing a wide range of audiences. Therefore, the methodology is designed to (1) identify broad thematic structures, (2) trace how specific policy domains evolve across election cycles, and (3) evaluate sentiment within those domains.

3.1 Data

The analysis draws on the U.S. political party platforms provided by the Manifesto Project Database (MARPOR), which supplies manifestos in the form of quasi-sentences.[25] These quasi-sentences are short semantic units, generally around 100 characters in length, designed to isolate discrete claims, propositions, or argument fragments. This representation format is widely used in comparative political text research because it minimizes heterogeneity in sentence length and reduces syntactic complexity, thereby facilitating topic modeling and other interpretive analyses. The corpus consists of nine documents: five Democratic and four Republican manifestos spanning the 2008, 2012, 2016, 2020, and 2024 elections. The GOP decided to reuse its platform from 2016 for the 2020 election, hence only four from Republican. All texts were processed at the quasi-sentence level to maintain consistency across years and across parties.

Although the MARPOR corpus provides standardized segmentation for most years, two documents required substantial preprocessing. The Democratic Party’s 2008 manifesto appeared in the dataset as a single unsegmented block of approximately 164,626 characters, in contrast to all other manifestos, which were pre-divided into quasi-sentences. Using this raw text would have resulted in distorted document embeddings and unreliable topic clustering, as the model would have treated the entire manifesto as a monolithic unit and forced it into a single dominant topic cluster. To ensure consistency, a custom quasi-sentence parser was developed. This parser employed punctuation-based rules for splitting long text blocks, supplemented by heuristics for common political discourse structures such as list markers, section boundaries, and connective phrases. Units shorter than 20 characters were removed to eliminate residual formatting fragments. The resulting segmentation approximates the MARPOR quasi-sentence style and aligns the Democratic 2008 text with later years.

The second issue arose from the formatting of the Republican Party’s 2024 platform, which differed significantly from all earlier Republican documents. The original text contained extensive chapter titles and section headings in full capital letters, table-of-contents lines interspersed with substantive content, non-breaking space characters (`\xa0`), and page numbers embedded in the text flow. When used without correction, these features produced several artifact-driven topic clusters composed primarily of formatting noise, thereby reducing interpretability and inflating topic counts. Because these anomalies were limited to the 2024 Republican manifesto, a targeted cleaning pipeline was implemented exclusively for this document. This pipeline removed all-caps structural markers, suppressed table-of-contents fragments, eliminated non-breaking spaces, stripped standalone page numerals, and normalized whitespace. Restricting these corrections to 2024 ensured that earlier Republican manifestos were not inadvertently altered.

In addition to these document-specific corrections, the full corpus underwent a general cleaning process designed to remove residual noise and ensure linguistic consistency. All quasi-sentences shorter than 20 characters were deleted, as were units with fewer than 70 percent alphabetic characters. Pure numerical strings, Roman numerals,

and structural or indexing artefacts were also filtered out. These steps mitigate the risk of meaningless units entering topic clusters or biasing embedding-based analyses.

Following preprocessing, the Democratic platforms contained approximately 1,400 quasi-sentences in 2008, 1,395 in 2012, 1,593 in 2016, 2,762 in 2020, and 2,900 in 2024, for a total of roughly 8,651 quasi-sentences. The average unit length across Democratic years ranged from roughly 95 to 114 characters, reflecting the increasing verbosity of recent platforms. The Republican platforms contained approximately 1,154 quasi-sentences in 2008, 1,793 in 2012, 2,289 in 2016, and 536 in 2024. The average unit length ranged from roughly 65 to 129 characters. The abrupt change in 2024 reflects a sudden change in style and structure, consistent with our empirical reading of the platform. These datasets provide sufficient granularity for topic modeling while preserving comparability across time.

3.2 Document Similarity

Before modelling the thematic structures of party platforms, the study first evaluates the overall semantic continuity of these documents over each election. Each document is first represented as a vector produced by a transformer model. The position of a document in the high-dimensional space reflects the semantic context of its language. Documents that use similar combinations of words to discuss related policy concerns are located in similar directions in the embedding space. In our study, we used the *bert-base-nli-mean-tokens* Sentence-BERT model because it is a widely used transformer architecture and already fine-tuned on natural language inference tasks.

Cosine similarity is then computed for all pairs of party platform documents. This metric evaluates the angle between these vectors, capturing similarity in semantic orientation rather than absolute magnitude. This makes it well suited for comparing political texts of different lengths, as it assesses whether platforms emphasize similar issue domains and conceptual framings rather than how much they say. The results are values between -1 and 1 that approximate semantic proximity, with -1 being completely unidentical and 1 being identical. This approach allows a direct comparison of content continuity within each party as well as cross-party dynamics across time.

This preliminary analysis serves two purposes. First, it identifies broad discontinuities that may signal meaningful rhetorical shifts. Second, it assesses whether topic-level divergence corresponds to genuine semantic movement at the document level. Document similarity, therefore, functions as an initial diagnostic, guiding the subsequent modeling stages.

3.3 Topic modeling

Having established patterns of continuity and change at the document level, the analysis turns to identifying the thematic structure that organizes platform content. To recover these broad policy areas, we use BERTopic, a topic model that integrates transformer embeddings, manifold learning, and density-based clustering. (Grootendorst 2022)[15]. The model inputs are quasi-sentences, the smallest meaningful units in the Manifesto Project’s annotation scheme. Each quasi-sentence is first encoded using Sentence-BERT, which maps short text units into dense vector embeddings that capture semantic meaning in context rather than treating words as independent tokens (Reimers and Gurevych 2019)[41]. In this representation, quasi-sentences that discuss related policy concerns using similar language patterns are positioned near one another in the embedding space, making them suitable inputs for clustering.

Because these embeddings reside in a high-dimensional space that is difficult to cluster directly, the embedding space is then reduced using UMAP. This is a nonlinear manifold learning method that can preserve both local structure and broader relational geometry (McInnes, Healy, and Melville 2018)[29]. This step helps the model to identify coherent semantic groupings while retaining meaningful distances between conceptually distinct issue areas.

Clusters are then identified using HDBSCAN. This is a density-based algorithm that groups quasi-sentences according to regions of high semantic density rather than forcing all observations into a fixed number of clusters (Campello et al. 2013)[4]. This property is particularly useful for this study, where thematic boundaries are often overlapping, issue salience varies across elections, and some policy domains are discussed much more extensively than others.

3.3.1 Stage 1: Broad Topics Discovery

Having defined how topics are constructed, the next step is to determine how many such clusters should be retained. Because BERTopic allows the number of topics (K) to be discovered automatically or set manually, the choice of K requires explicit justification. The default unsupervised clustering outcome produced a large number of highly granular clusters. Hence, we estimated a sequence of candidate BERTopic models with different values of K and evaluate each model using two complementary metrics: topic coherence and topic diversity. Coherence, measured using the c_v metric, assesses the semantic consistency of the most representative words within each topic. Diversity measures the proportion of unique top words across topics and thus penalizes redundant or overly similar clusters. Models that produce high coherence but low diversity tend to overfit by splitting similar concepts into multiple tightly clustered topics, whereas models with high diversity but low coherence produce overly broad, semantically weak topics.

Figure 1 plots coherence and diversity scores across candidate values of K, illustrating the trade-off between interpretability and over-fragmentation. Coherence improves notably between K=30 and K=35, while diversity remains sufficiently high to avoid collapse into redundant clusters. Beyond this range, coherence stabilizes and diversity declines, indicating diminishing returns. Using this joint criterion—maximizing coherence while maintaining high topic diversity, we selected K=35 for the corpus as a parsimonious and substantively meaningful solution. Fixing the number of broad topics enables consistent comparison of thematic prevalence across election cycles.

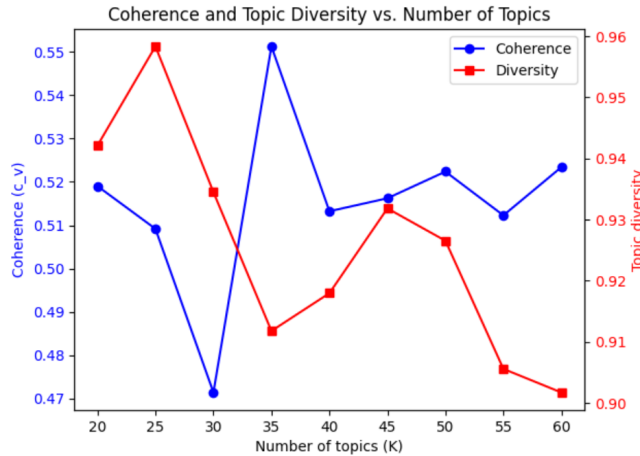


Figure 1: Coherence and Topic Diversity Scores vs Number of Topics

Once the optimal number of topics is determined, the model is fitted to the full multi-year corpus. Each quasi-sentence receives a topic label, and the distribution of topics within each election cycle is computed by aggregating the share of quasi-sentences assigned to each cluster. This produces a year-by-topic matrix reflecting thematic prevalence over time. Changes between consecutive elections are calculated to measure how the party’s policy emphasis evolves. To aid interpretation, the analysis includes heatmaps of topic intensities, transition plots showing the magnitude and direction of thematic shifts, and summary tables detailing the most distinctive words associated with each topic.

3.3.2 Stage 2: Subtopic Discovery within Broad Topics

Broad topics often contain multiple conceptually distinct subthemes. For instance, a broad immigration topic may encompass legal immigration, border enforcement, refugee policy, and labor market concerns. Similarly, technology-oriented topics may combine digital infrastructure, cybersecurity, data privacy, and artificial intelligence governance. To investigate such internal heterogeneity, we performed a second layer of modeling within three policy domains of substantive interest: immigration, healthcare, and technology.

Using the broad-topic assignments from Stage 1, I extract all quasi-sentences associated with each selected topic for each election year. For each topic-year pair, we then fit a dedicated BERTopic model with parameters tuned for finer granularity, allowing the recovery of subtopics that capture distinct framings within the broader issue area. Comparing subtopics across consecutive elections reveals how internal components of an issue evolve over time—for example, whether discussions of technology shift from broadband access and innovation policy toward

algorithmic accountability and AI safety. Subtopic-word distributions, prevalence estimates, and word-importance grids provide both qualitative and quantitative insight into these transformations.

3.4 Sentiment Analysis

To complement the structural analysis with an examination of tone, the final stage applies transformer-based sentiment classification to quasi-sentences in the selected policy areas—immigration, healthcare, and technology. Whereas the previous stages identify what issues are emphasized and how these issues are internally structured, the sentiment analysis focuses on how they are framed rhetorically. This shift from thematic structure to evaluative language provides a fuller account of the party’s changing communication style.

Sentiment labels are derived using the siebert/sentiment-roberta-large-english model, a RoBERTa-large architecture fine-tuned on a large-scale corpus for sentiment analysis. This model is widely used in applied NLP research due to its strong performance and its capacity to handle formal, policy-oriented text more reliably than models trained primarily on conversational sources such as social media. For each quasi-sentence, the classifier produces both a categorical label (positive or negative) and a confidence-weighted probability score, which is transformed into a continuous value ranging from -1 to 1 . These signed scores capture graded shifts in sentiment intensity, allowing for smoother temporal comparisons across election cycles.

To ensure the robustness of year-level sentiment profiles, the analysis excludes topic-year combinations with insufficient observations, as very small sample sizes tend to generate unstable or extreme sentiment estimates. Mean sentiment values are then computed for each policy domain in each election year. These aggregated measures reveal whether the party’s evaluative framing of key issues becomes more positive, negative, or neutral over time, and whether these tonal shifts correspond to the internal thematic changes identified in the subtopic modeling stage. By linking structural and affective dimensions of the text, the sentiment analysis completes the progression from broad document similarity to fine-grained rhetorical characterization.

4 Results

4.1 Document Similarity

The document similarity analysis quantified the textual overlap between party platforms in election cycles (see Figure 2). For the Democratic Party, pairwise similarity scores fluctuated throughout the observation period. The similarity between the 2008 and 2012 platforms was 0.612, while the highest similarity was observed between the 2012 and 2016 platforms (0.852). Comparisons involving the 2024 Democratic platform yielded the lowest coefficients in the data set, ranging from a minimum of 0.485 (vs. 2012) to a maximum of 0.688 (vs. 2008). In contrast, Republican Party platforms consistently displayed similarity scores within the range of 0.776 to 0.809 for all distinct manifestos. The analysis also identified a cosine similarity of 1.000 between the Republican 2016 and 2020 platforms, indicating that the party utilized the exact textual content in two election cycles without making any changes.

4.2 Topic modeling

From the 35 latent topics initially extracted for each party, the top 20 most variable topics were selected for visualization to capture the most significant programmatic shifts (Figures 3 and 4). The observed trends reveal two distinct categories of results: party-specific policy domains that exhibit unique temporal trajectories, and overlapping thematic areas present in both platforms. The following section first details the distinct topic evolution within the Democratic and Republican platforms individually, highlighting party-specific priorities. Subsequently, the analysis addresses shared topics—such as healthcare, immigration, and technology to compare how each party’s emphasis on these common issues evolved over the observed election cycles.

4.2.1 Major topics in Democrat’s platform

The topic distribution for the Democratic Party (see Figure 3) revealed a set of enduring policy priorities alongside significant shifts in emphasis over time. The topic of “human rights and voting rights” maintained a consistent presence throughout the observation period, fluctuating narrowly between 8.7% and 11.9% across all five election cycles. Similarly, “healthcare” remained a central theme, peaking at 12.5% in 2020 before settling at 8.9% in 2024. In contrast, the prominence of the “nuclear weapons and foreign policy” topic exhibited a notable decline; it was

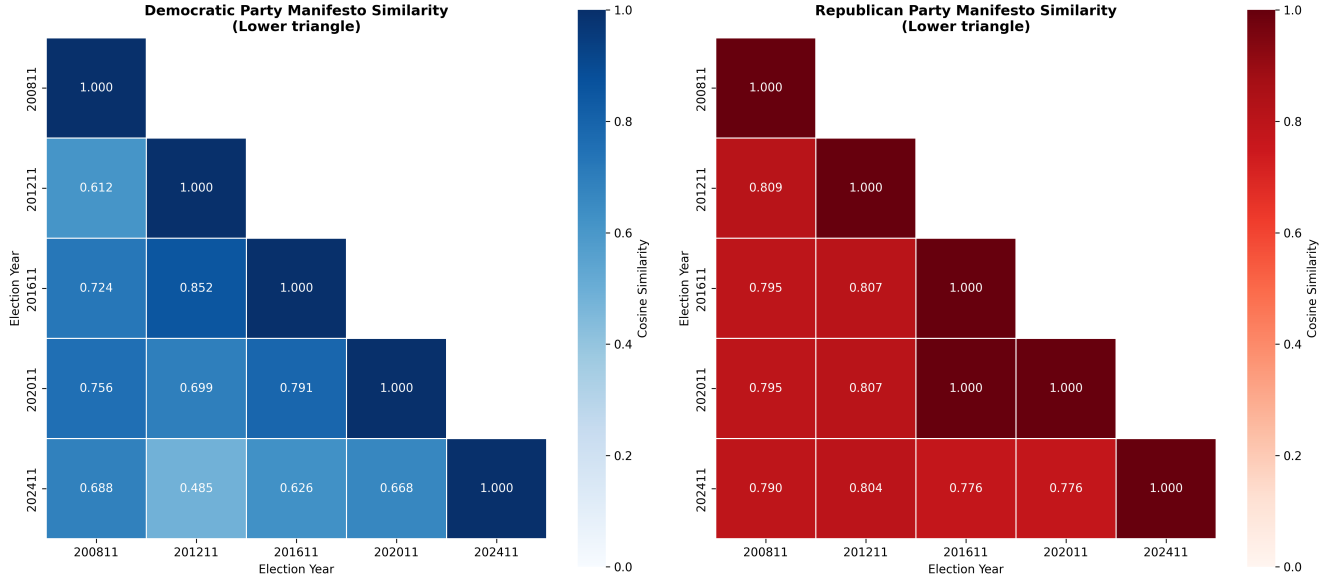


Figure 2: Document similarities across party platforms for Democrats and Republicans

the most prevalent topic in the earlier cycles (16.3% in 2008 and 20.0% in 2012) but dropped to approximately 8.0% in the three subsequent elections. Economic topics, specifically those relating to the “middle class” and “workers rights,” showed moderate but stable engagement, with the middle-class topic peaking at 5.1% in 2012 and declining to 2.0% by 2024.

Emerging domains also demonstrated distinct trajectories in the later election cycles. Topics related to “climate change” and “clean energy” showed a gradual increase in prevalence, rising from 5.9% in 2012 to 8.7% in 2024. Technology-related issues became increasingly visible; while “AI safety” was negligible in earlier platforms (< 0.1%), it emerged with a 1.2% prevalence in 2024. The 2024 cycle also introduced specific new focuses, most notably the topic associated with “Vice President Harris and executive orders,” which rose to 7.4%, significantly higher than in any previous year. Conversely, the “discrimination and civil rights” topic saw a sharp temporary spike to 12.5% in 2020 before returning to 4.1% in 2024, mirroring the fluctuation patterns seen in other responsive domains.

4.2.2 Major topics in Republican’s platform

The topic distribution for the Republican Party (see Figure 4) highlighted significant volatility in platform emphasis, particularly in the 2024 election cycle. The topic labeled “current administration” exhibited the most pronounced shift, rising to a prevalence of 17.8% in 2024 after remaining within the 3.3%–5.6% range during the 2008–2020 period. Similarly, the topic regarding “immigration and secure borders,” which fluctuated between 4.9% and 6.8% in prior cycles, increased to 11.3% in 2024. Other domains also saw increased salience in the final observation year: “higher education” rose to 8.9% (compared to a prior maximum of 4.7%), and the topic concerning “judges” reached 7.1%, up from an average of approximately 3.4% in previous years. Additionally, the topic of “inflation” emerged with a prevalence of 3.3% in 2024, having been negligible (< 0.7%) in all prior cycles.

In contrast, several topics that were dominant in earlier platforms showed a marked decline by 2024. The “Israel, NATO, and Middle East” topic, which peaked at 11.5% in 2008, decreased to 4.2% in the most recent cycle. Likewise, the “Fifth Amendment” topic, which consistently held a prevalence above 8.0% from 2008 through 2020 (peaking at 11.8%), dropped to 4.7% in 2024. Topics related to “restructuring tax” and “Obamacare/abortion” followed similar downward trajectories; tax-related mentions fell from 8.9% in 2008 to 3.3% in 2024, while the abortion and healthcare funding topic declined from 9.5% to 4.5% over the same period. Finally, topics related to “economic growth” were present throughout the period but fluctuated in intensity, peaking at 8.2% in 2012 before settling at 5.6% in 2024.

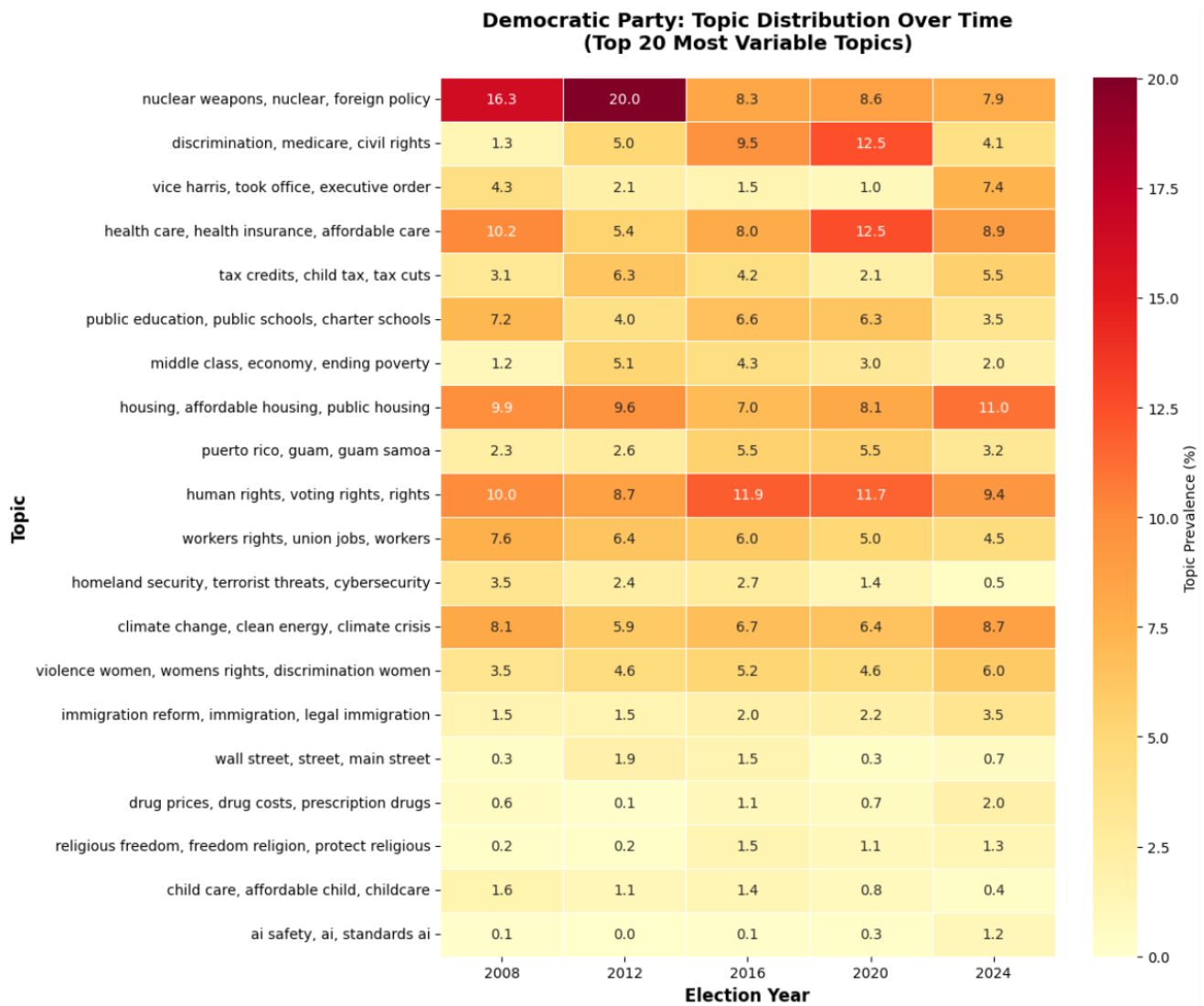


Figure 3: Democrat topic prevalence over time

4.2.3 Shared topics across the two parties

While the topic modeling analysis highlighted distinct programmatic priorities for each party, it also identified three key policy domains that emerged as significant features in both the Democratic and Republican datasets: healthcare, immigration, and technology. This overlapping presence permitted a direct comparative analysis of topic prevalence and temporal evolution across the two platforms. The following results detail the divergent trajectories and fluctuating emphasis observed within these shared thematic areas.

Healthcare

The Democratic healthcare subtopics (see Figure 5) increase and decrease in volume across cycles and vary in their internal composition. In 2008, the platform includes several subtopics covering care access, insurance coverage, disability-related issues, research, and infectious disease. In 2012, the total is slightly higher, with quasi-sentences distributed among insurance coverage, Medicaid and Medicare, disability services, infectious disease treatment, and related areas. In 2016, the number rises further, and the subtopics include mental health, HIV/AIDS, disability-related care, and general healthcare access. In 2020, the total volume increases substantially, with content spread across COVID-19, research centers, treatment access, disability services, and related categories. In 2024, the number decreases, and the quasi-sentences are distributed among health care, insurance coverage, new research initiatives, disability-related issues, and country-level health concerns. The composition differs noticeably across years, with the size and mixture of subtopics changing in each cycle.

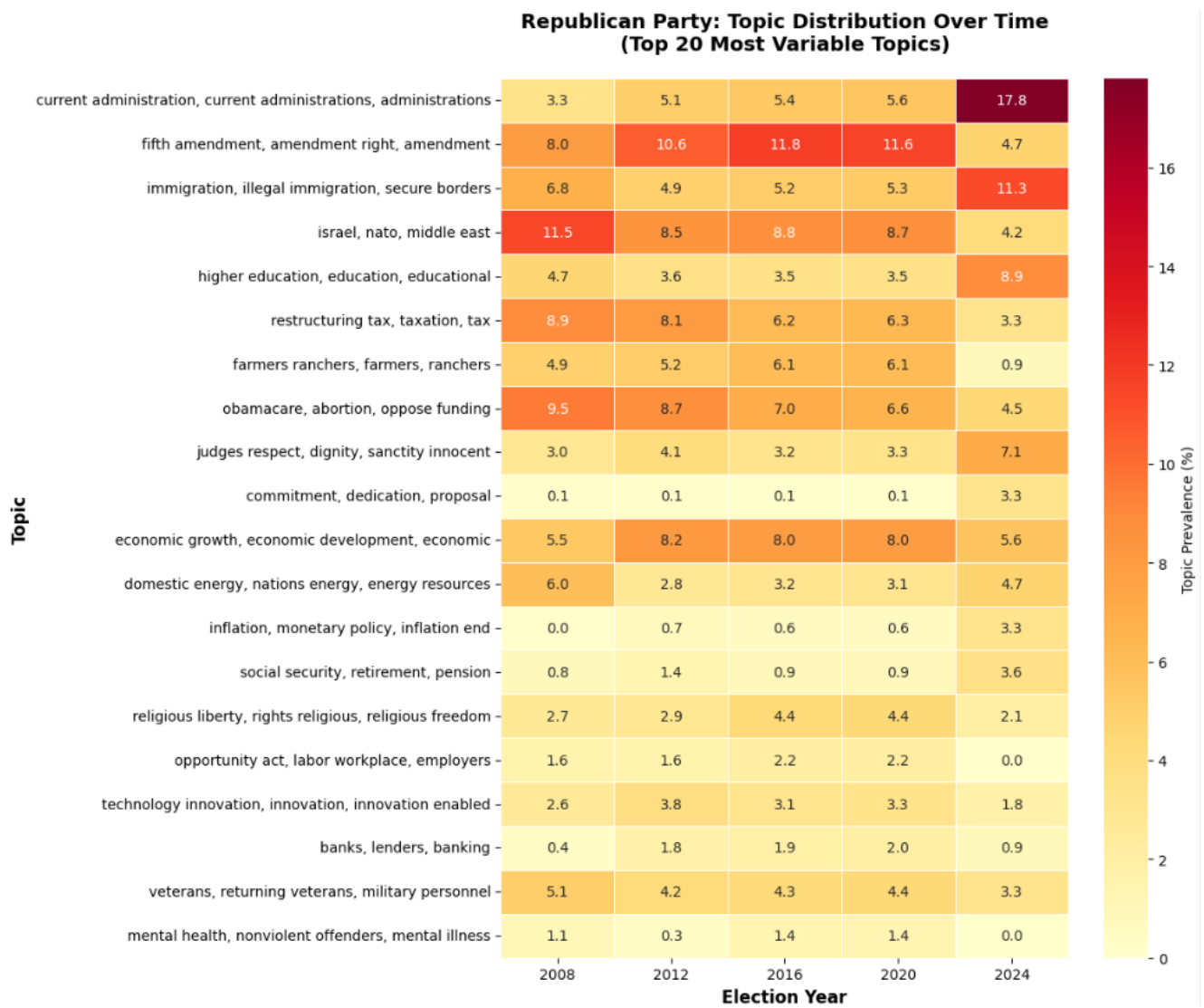


Figure 4: Republican topic prevalence over time

Republican healthcare subtopics (see Figure 6) display large fluctuations in total quasi-sentence volume. In 2008, the platform contains a moderate number of healthcare-related quasi-sentences divided among coverage for families, Medicaid/Medicare competition, and concerns related to legal suits. In 2012, the number increases sharply, with the majority devoted to insurance costs and additional subtopics including Medicaid and Medicare programs. In 2016, the volume decreases but remains high, and the subtopics are distributed across Medicare and Medicaid, ACA-related terms, and affordable care. In 2020, the total declines further, with quasi-sentences primarily focused on Medicaid and the ACA. The distribution in each year reflects a distinct arrangement of subtopics, and the relative weights of each subtopic vary substantially across cycles.

Immigration

Democratic immigration subtopics (Figure 7) show distinct distributions across election years. In 2008, the platform contains a small number of quasi-sentences divided between broad immigration reform and concerns related to immigrant communities. In 2012, the volume is slightly lower, with subtopics covering deportation, security, and general immigration-needs statements. In 2016, the amount increases and is distributed among refugees, broken immigration systems, and immigrant unification themes. In 2020, the total grows considerably, with quasi-sentences organized across asylum seekers, immigration in the 21st century, family-based green-card processes, and general immigration-system descriptions. In 2024, the volume decreases slightly, and the subtopics include migration security and refugee reform. Each year presents a different set of subtopics, and the structure changes substantially

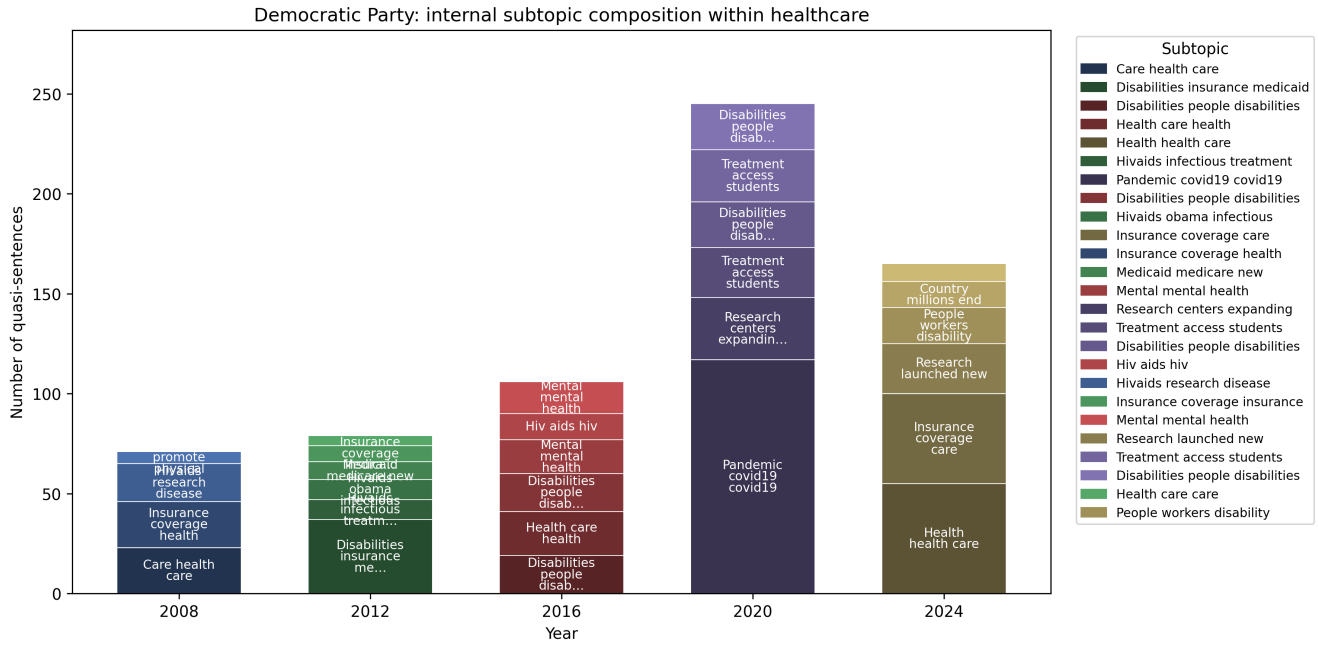


Figure 5: Democrat healthcare subtopic distribution

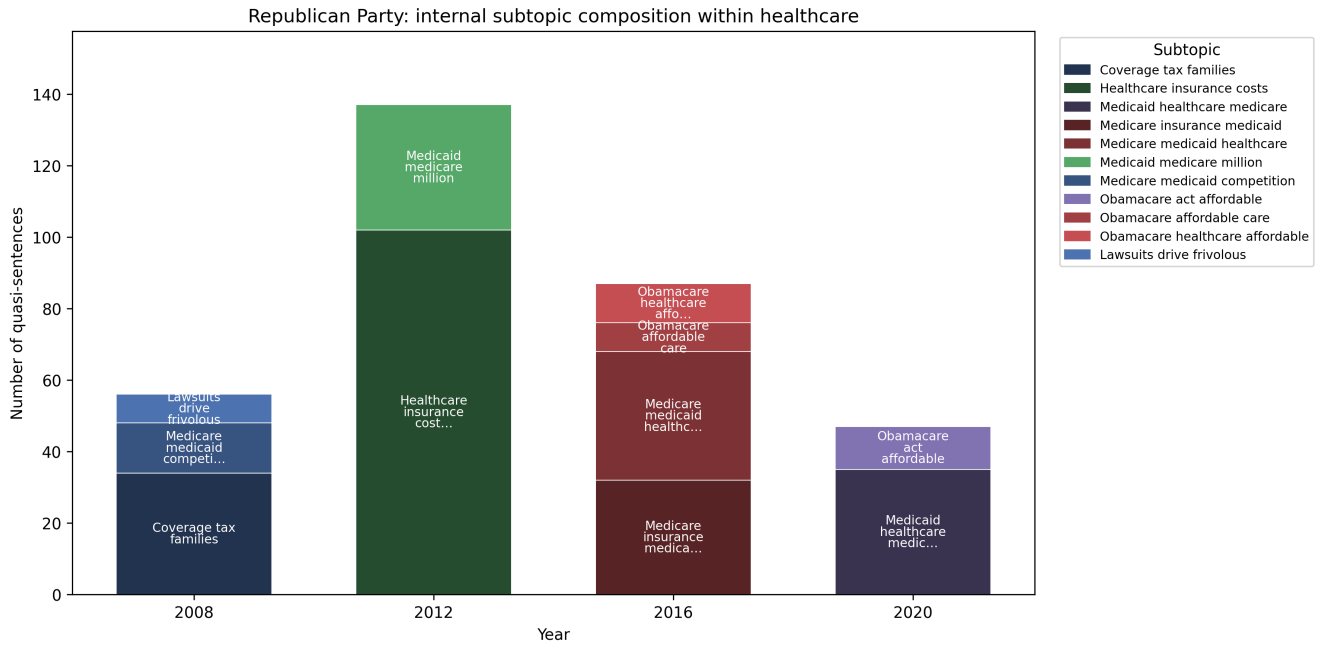


Figure 6: Republican healthcare subtopic distribution

over time. The internal composition of Republican immigration subtopics (Figure 8) shifts considerably across cycles. In 2008, the platform features several subtopics covering illegal immigration, immigrant communities, and refugee acceptance. In 2012, the quasi-sentences are grouped around immigration rules, illegal immigration, and assistance for new immigrants. In 2016, the vast majority of immigration-related quasi-sentences fall under a single subtopic concerning legal borders and immigration. In 2020, the distribution broadens again, with subtopics including border entry, immigration and citizenship, deportation, asylum, and legality. In 2024, the total number decreases, with the content divided between border concerns and illegal immigration enforcement. The number and type of subtopics differ strongly from cycle to cycle, and no single pattern persists across years. **Technology**

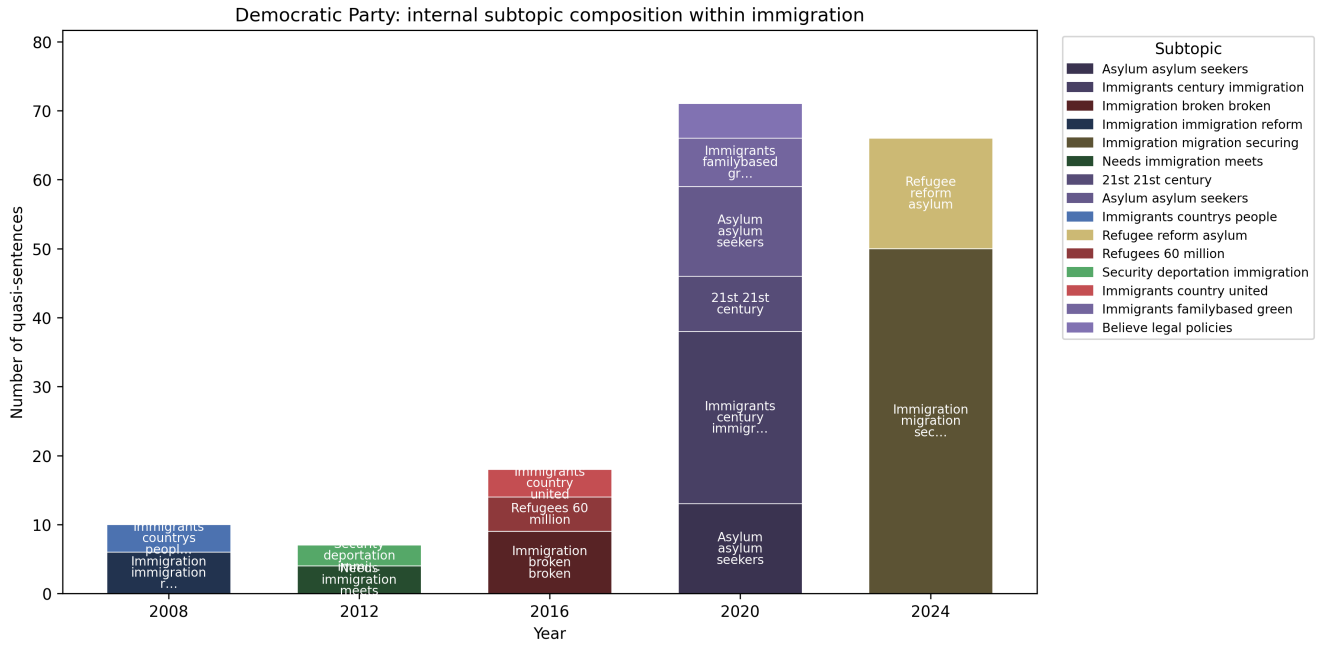


Figure 7: Democrat immigration subtopic distribution

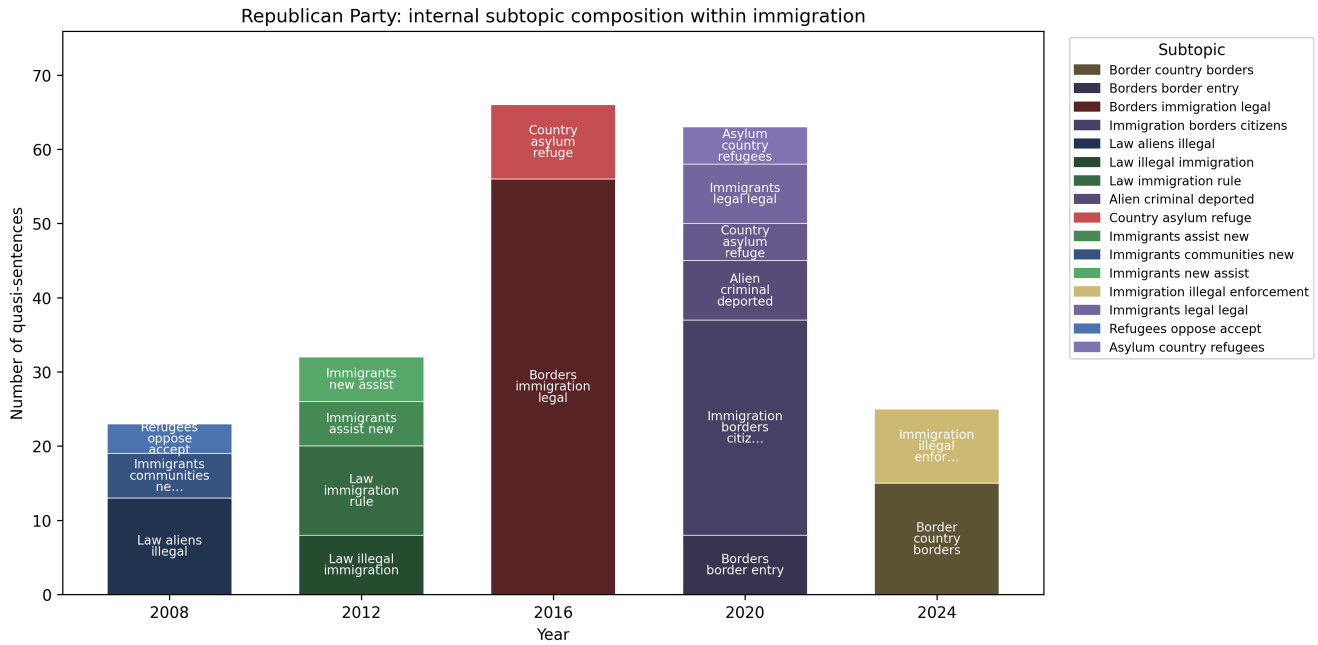


Figure 8: Republican immigration subtopic distribution

The Democratic privacy and data-rights subtopics (see Appendix Figure 8) appear only in the 2020 and 2024 platforms. In 2020, the quasi-sentences are divided between privacy-rights protection, broadband access, and internet-access-related issues. In 2024, the number increases, and the distribution consists primarily of data-privacy protection and social-media-related subtopics. Across the two years, the total volume and the relative share of each subtopic differ, and the internal composition shifts between infrastructural concerns and privacy-specific language.

The Republican technology subtopics (see Appendix Figure 11) show substantial variation in both the number of quasi-sentences and the composition of subtopics across election years. In 2008, the platform contains a small set of technology-related quasi-sentences, primarily divided between information technology and general innovation

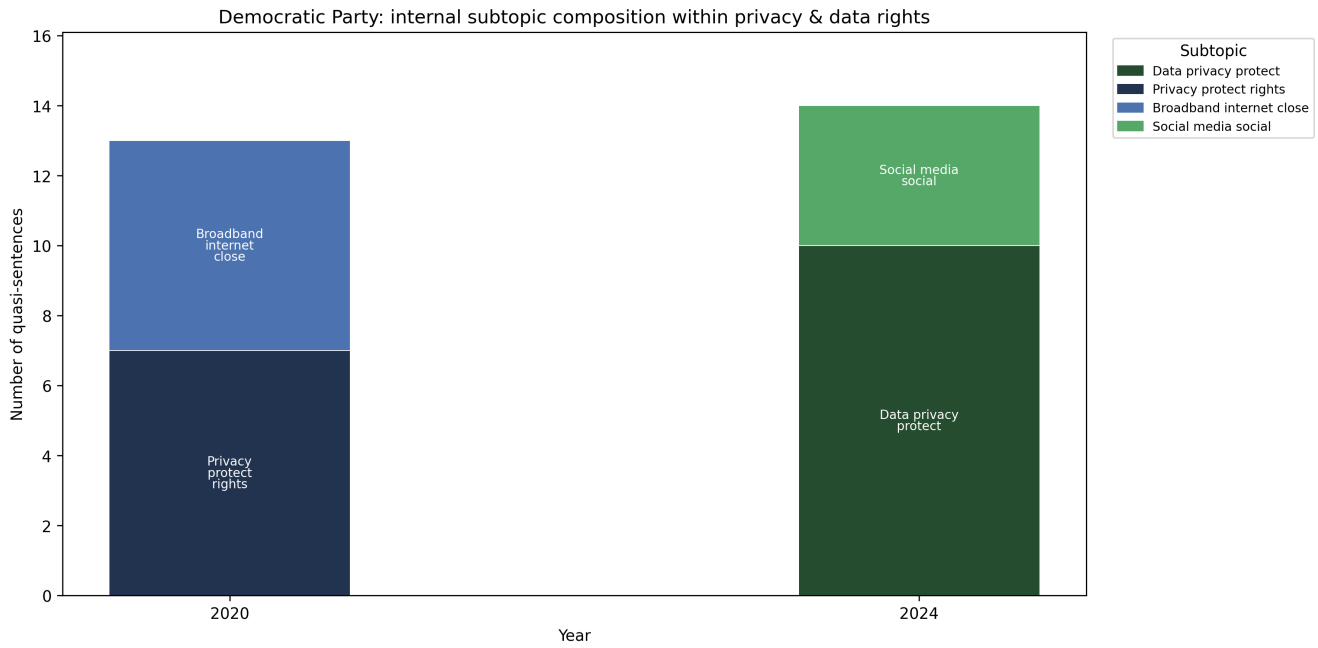


Figure 9: Democrat technology subtopic distribution

themes. In 2012, the volume increases noticeably, and the distribution includes multiple subtopics such as telecommunications technology, telecommunications commissions, science-related themes, and local state issues. In 2016, the total volume decreases again, with subtopics concentrated in innovation access, innovation growth, technological medical themes, and technological learning. By 2020, the number of quasi-sentences is the smallest among the four years shown, with only two subtopics—innovation growth and technological advancement—accounting for the total. Overall, the internal composition varies substantially from cycle to cycle, with different subtopics appearing and disappearing across years.

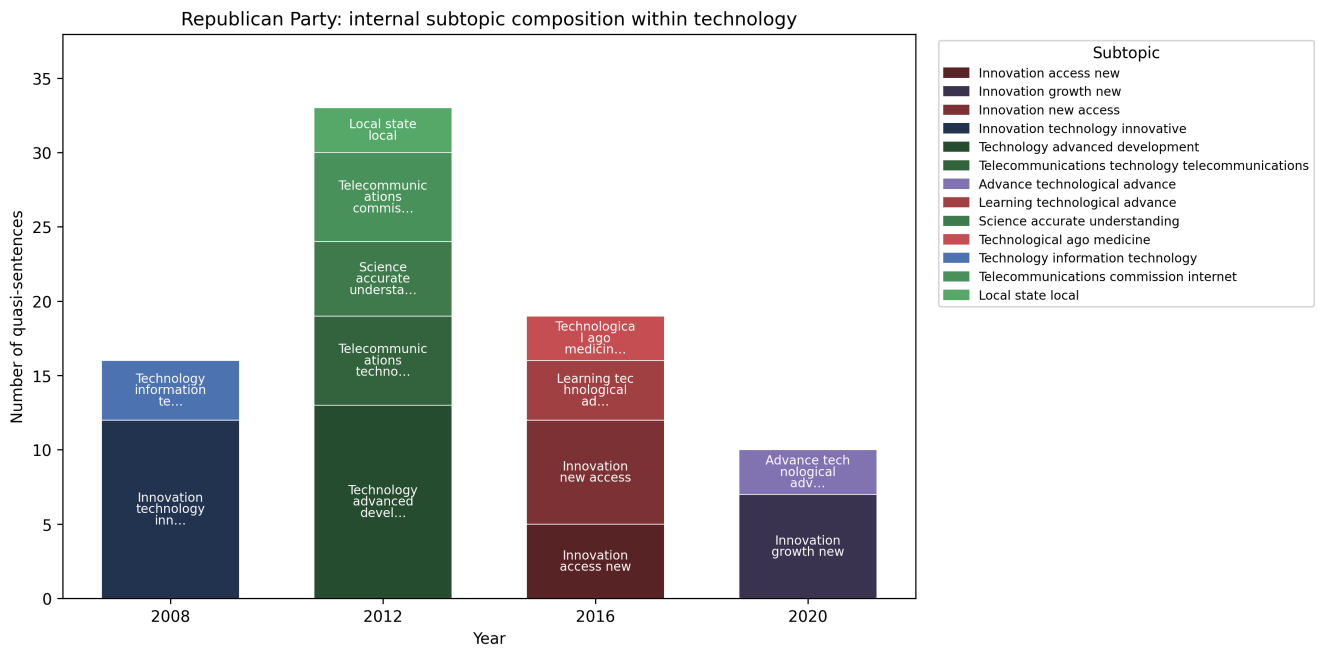


Figure 10: Republican technology subtopic distribution

4.3 Sentiment analysis

The sentiment analysis examines the emotional tone associated with policy discussions across healthcare, immigration and technology (see Figure 11). The results reveal substantial differences between the two parties over time, both in the polarity of sentiment within issue domains and in the trajectories across election cycles.

Healthcare

Democratic platforms consistently display a highly positive sentiment profile in healthcare, with most years exhibiting strong positive polarity and very few negative statements. The 2008, 2012 and 2016 platforms all show positive sentiment concentrated around access, affordability and protection of rights. Sentiment remains mostly positive in 2020 and 2024, although there is a slight decrease in the proportion of very positive statements in 2024. Across all cycles, negative sentiment appears only minimally in Democratic healthcare discussions, indicating that the party consistently frames healthcare in supportive and expansive terms.

In contrast, Republican healthcare sentiment is consistently more negative. Across all cycles, the largest share of Republican healthcare statements falls into the negative category, with only a small proportion coded as positive. This pattern remains stable from 2008 through 2024. Even in years when positive sentiment increases slightly, negative sentiment still dominates Republican healthcare messaging. These results suggest that the two parties maintain distinct rhetorical orientations toward healthcare, with Democrats emphasizing positive expansions and Republicans emphasizing critiques and perceived risks.

Immigration

Democratic sentiment on immigration shows greater fluctuation across election cycles. In 2008 and 2012, immigration sentiment is moderately positive, reflecting emphasis on reform and inclusion. Sentiment becomes more negative in 2016, largely due to increased references to enforcement challenges. In 2020, the distribution remains mixed, with both positive and negative components. By 2024, sentiment shifts positive again, driven by language focused on rights, pathways to citizenship and protections for migrants. Overall, Democratic sentiment on immigration is variable but trends toward a more positive framing in 2024.

Republican sentiment on immigration is strongly and consistently negative across all cycles. The vast majority of immigration-related statements are coded as negative, with little variation between elections. Positive sentiment is nearly absent throughout the observation period. This pattern indicates a stable rhetorical focus on enforcement, border security and perceived threats. Compared to healthcare, the differences between the two parties on immigration are even more pronounced.

Technology

Democratic sentiment toward technology is predominantly positive in most cycles. The 2008, 2012 and 2016 platforms all reflect optimism about innovation, digital rights and infrastructure. Sentiment becomes slightly more mixed in 2020, with references to regulatory concerns and inequality. In 2024, sentiment becomes strongly positive again, driven by discussions of innovation, AI safety and digital equity. Negative statements remain limited across all years.

Republican sentiment toward technology is more mixed overall. In 2008, sentiment skews positive, reflecting support for innovation and competitiveness. In 2012 and 2016, sentiment becomes more neutral, with fewer clearly positive statements. In 2020 and 2024, sentiment does not shift strongly in either direction, resulting in a balanced but less emotionally charged profile compared to the Democrats. Across the entire period, Republicans show neither the consistently positive tone of the Democrats nor the strongly negative pattern observed in immigration.

Across all three domains, the sentiment results indicate that the Democratic Party generally employs more positive evaluative language, although with variation depending on the issue and year. The Republican Party, by contrast, shows consistently negative tone in immigration, consistently negative tone in healthcare and a more neutral or mixed tone in technology. These patterns reveal significant differences in how the two parties emotionally frame policy discussions

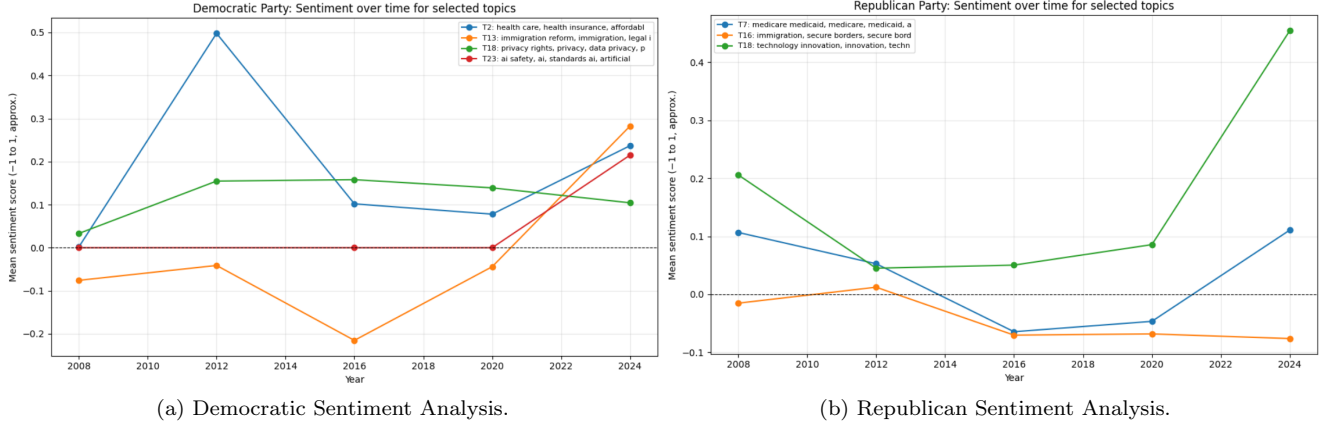


Figure 11: Sentiment Analysis for Democratic and Republican Party

5 Discussion

5.1 The evolution of party agendas

The similarity matrices (Figure 2) offer the first indication that partisan divergence is not merely a result of increasing ideological distance, but of structurally different trajectories of agenda construction. Democratic platforms exhibit moderate internal continuity from 2008 to 2024, apart from sharp drops in similarity between 2012 and 2016 and again between 2020 and 2024. These declines have coincided with major exogenous shocks and issue innovations, including the post-Affordable Care Act reorganization of domestic policy, the COVID-19 pandemic, and the emergence of platform governance and AI safety. This pattern is consistent with research showing that Democrats adjust agenda structure and policy language more responsively to emerging governance challenges and crisis events (Grossmann and Hopkins, 2015[17]; Cohen et al., 2023[7]).

By contrast, Republican platforms display far greater stability across election cycles. Prior work characterizes Republican messaging as organized around a durable architecture centered on constitutional liberties, enforcement, and national sovereignty (Noel, 2013[35]; Mettler and Lieberman, 2020[30]). Our similarity results support this account: new issue language is typically incorporated into preexisting frames rather than prompting broader structural reorganization of the agenda.

Together, these patterns reinforce findings on asymmetric party evolution, in which Democratic agendas change through issue uptake and reframing, while Republican agendas emphasize message stability, consistency, and the amplification of long-standing priorities (Grossmann and Hopkins, 2020[16]). These asymmetries set the stage for the more granular divergences observed in the topic-model heatmaps.

5.2 Topic distributions and agenda divergence

The heatmaps (Figures 3 and 4) suggest that the two parties are not merely advancing different policy solutions within a shared agenda, but increasingly place emphasis on different sets of issues when constructing their governing platforms. Across election cycles, Democratic platforms tend to devote greater attention to healthcare, anti-discrimination protections, housing, workers’ rights, climate change, and, in later years, data rights and AI governance. Republican platforms, by contrast, devote relatively more attention to immigration and border enforcement, constitutional liberties and property rights, energy production, religious liberty, and foreign policy in the Middle East.

These patterns are consistent with what scholars describe as “agenda polarization”: divergence not only in policy preferences but in issue salience and selection (Hare and Poole, 2014[18]; Barber and McCarty, 2015[2]). Research on U.S. platform evolution shows that parties increasingly assemble policy agendas around distinct clusters of issues aligned with the identities and priorities of their core coalitions (Pomper and Lederman, 2019[40]). The heatmaps provide empirical evidence of this tendency, with Democratic agendas emphasizing welfare-state expansion, rights, and regulatory capacity, and Republican agendas emphasizing sovereignty, enforcement, and market-oriented governance.

As a result, the two platforms are not just at odds in terms of policy, but also diverge in the mix of issues they emphasize. This kind of polarization in political discourse makes bipartisan cooperation more difficult, since it narrows the range of policy challenges that both parties are willing to address together, even when their interests overlap.

5.3 Issue Divergence within Shared Domains

Where the parties do address the same broad domain, the internal structure of topics and subtopics differs substantially. These differences in semantic clustering indicate distinct problem definitions and governance logics.

5.3.1 Health care

Both political parties dedicate significant attention to health care, but their approaches are rooted in fundamentally different problem definitions, leading to distinct patterns in their policy subtopics. Democratic health-care subtopics show a tendency to broaden over time (Appendix Figure 9 and 12). Initially centered on insurance and coverage (2008–2012), they expand to encompass themes like mental health, disability rights, and public health research (2016), before being temporarily dominated by a COVID-19–related cluster in 2020, and subsequently returning to a focus on general health-care access, insurance coverage, and long-term obligations in 2024. The sentiment surrounding the Democratic health topic is consistently positive, peaking around 2012 and indicating a framing of health care as a positive rights and capacity-building project—expanding access, investing in public health, and treating pandemic response as extensions of this commitment. This pattern aligns with existing political science work demonstrating that left-of-center parties interpret health systems through a positive-rights and public-capacity lens (Patashnik and Zelizer, 2020[39]).

By contrast, Republican health-care subtopics consistently revolve around costs, entitlement reform, and opposition to the Affordable Care Act (ACA) across all electoral cycles. Early platforms favored market-based solutions like tax-favored coverage; 2012 and 2016 platforms were heavily focused on criticizing the cost and structure of “Obamacare”; and 2020 continued to pair Medicaid/Medicare reform discussions with ACA opposition. The sentiment within the Republican topic turns mildly negative around 2016–2020, suggesting a shift from promising market alternatives to criticizing existing policy, only returning to slight positivity by 2024. This conservative emphasis on government overreach, fiscal burdens, and regulatory distortion is well-documented in research on conservative health policy (Oberlander, 2010[36]; Béland et al., 2016[3]).

Polarization in health care is not just about the magnitude of government spending but more about the fundamental definitions of the problem: Democrats see insufficient coverage and public-health capacity as the central issue, while Republicans highlight fiscal burdens and governmental distortion of the market.

5.3.2 Immigration

Immigration yields the clearest and sharpest in-topic divergence, a pattern strongly consistent with the broader literature on polarized immigration rhetoric. On the Democratic side, the internal structure of the immigration topic shifts over time, moving from a focus on legal pathways and “immigration reform” (2008–2012) to foregrounding refugees and global humanitarian crises (2016). By 2020, the focus shifts toward asylum seekers and system modernization, and by 2024, the platform settles into a dual structure focusing on managing migration and border processes alongside a separate refugee-reform component. Sentiment within the Democratic topic starts slightly negative (2008–2012) but moves strongly positive by 2024, emphasizing contributions and moral obligations alongside recognition of system capacity. This reflects an increasing adoption of a humanitarian, inclusion-oriented frame.

Republican immigration subtopics trace an almost opposite trajectory. In 2008–2012, the language mixed talk of illegality (“law, aliens, illegal”) with some community integration talk. However, from 2016 onward, the structure converges almost entirely on border-centric clusters, dominated by language related to “borders,” “security,” and “alien criminal deported” in 2020 and 2024. The sentiment series for the Republican topic was near zero in 2008–2012 but became consistently negative from 2016 through 2024. This stark contrast implies a move toward a threat- and enforcement-oriented framing by Republicans, presenting immigration as primarily a security and sovereignty problem. This shift in both topic content and emotional tone mirrors well-documented partisan realignment in immigration rhetoric (Fowler et al., 2021[11]; Hopkins et al., 2022[22]), showing that Democrats increasingly frame the issue positively and Republicans negatively.

5.3.3 Technology, privacy, and data rights

The topic of Privacy and Technology offers a distinct form of partisan divergence: both parties express general support for technology, but their specific focus and resulting policy concerns differ significantly. On the Democratic side, a distinct privacy/data-rights topic appears only in 2020, initially combining consumer privacy with addressing the digital divide. By 2024, the internal structure evolves to focus on "data privacy protect" and "social media" subtopics, signaling a growing concern with platform governance, personal data, and the social consequences of digital platforms. At the same time, a new AI-safety topic emerges with clear, mildly positive sentiment by 2024, reflecting optimism about regulation and standard-setting as protective measures. This progressive policy focus on externalities, rights, and oversight is documented in U.S. tech politics literature (Gorwa, 2019[12]; Metz and Remond, 2023[31]).

In contrast, Republican platforms discuss technology within a broader topic dominated by "innovation," "advanced development," and sectoral growth (telecommunications, medicine, science). While subtopic presence rotates through different applications, there is little evidence of standalone data-rights or AI-governance clusters. Sentiment is positive throughout and becomes strongly positive by 2024, suggesting enthusiasm for technology primarily as an engine of economic growth, competitiveness, and national strength.

As a result, the divergence in technology-related framing appears as a governance vs. growth debate: Democrats increasingly frame technology around regulation, rights, and cross-cutting social impacts (privacy, social media, AI safety), while Republicans focus on innovation, economic opportunity, and competitiveness.

5.4 Sentiment trajectories and emotional polarization

The sentiment patterns observed across the three policy domains align closely with established theories of issue ownership and party communication strategy. Prior research suggests that parties tend to employ more positive language when discussing issues over which they claim institutional competence or moral authority. They tend to use affirmative language to signal credibility, feasibility, and alignment with core values (Walgrave et al., 2012[47]). In this context, Democrats' consistently positive sentiment in discussions of healthcare, climate change, and social rights reflects their longstanding association with welfare-state expansion, regulatory capacity, and inclusive social policy. Positive evaluative language frames these domains as areas of achievable progress and moral obligation, reinforcing the perception that government action is both legitimate and effective.

Republican sentiment patterns differ systematically. Negative sentiment is especially pronounced in domains such as immigration and abortion, consistent with research showing that conservative political communication more frequently relies on threat, decline, and loss framing to mobilize support and signal urgency (Clifford and Jerit[6], 2013[6]; Nai, 2021[34]). Rather than indicating simple opposition, negative language in these contexts often serves to construct policy domains as sources of risk, disorder, or moral transgression requiring containment or reversal. The surge in negative sentiment observed in Republican platforms during the 2012 and 2016 cycles aligns with broader rhetorical shifts associated with Tea Party mobilization and the emergence of insurgent populist appeals in the early Trump era, when emotional intensity became a central feature of partisan messaging.

Taken together, these divergent sentiment trajectories contribute to a widening emotional gap between the parties, echoing findings in the literature on affective polarization. In such environments, political conflict becomes structured not only around policy disagreement but around competing emotional narratives that define how issues are experienced and evaluated (Iyengar et al., 2019[23]; Druckman and Levendusky, 2019[9]). The partial moderation of Republican sentiment in 2024 and the pronounced positive shift in Democratic framing of abortion following the Dobbs decision suggest that emotional tone remains responsive to changing institutional and agenda-setting conditions. These dynamics underscore that sentiment is not merely a stylistic feature of party platforms, but an integral component of how parties construct political meaning and mobilize support.

5.5 Limitations and Further Work

Several limitations should be acknowledged with this project. Topic models provide a useful way of identifying clusters of co-occurring language, but they do not map directly onto predefined policy concepts. The boundaries between topics are therefore approximate. They capture broad semantic tendencies rather than the precise categories that parties might use internally. It is important to recognize that shifts in topic prevalence should be interpreted as changes in emphasis rather than exact measures of agenda allocation.

The subtopic analysis introduces similar considerations. BERTopic identifies finer-grained clusters, and many of these align well with recognizable sub-themes within healthcare, immigration and technology. However, the model can sometimes combine rhetorically distinct ideas or, conversely, fragment what appears to be a single coherent argument. Future work could compare multiple clustering approaches to validate the subtopic structure more systematically. This would help distinguish algorithmic artifacts from substantive patterns in party communication.

Sentiment analysis adds another layer of complexity. Although transformer-based models perform well on general sentiment tasks, they can misinterpret political rhetoric, especially when negative tone is used to signal urgency and threat or moral framing rather than disapproval. This study treats sentiment as a descriptive indicator rather than an interpretive endpoint, but future research could explore domain-specific models trained on political corpora or fine-tune existing models on annotated platform text. Such adaptations could offer a more precise understanding of how emotional tone functions within partisan messaging.

Another limitation is that platforms represent only one component of party communication. They are intentionally crafted, relatively formal documents that reflect negotiation among party actors. Their tone and content differ from speeches, debates, campaign advertisements or social media messaging, which may be more reactive or more polarized. Expanding the analysis to other genres would allow researchers to assess whether agenda-level divergence is consistent across communication modes or whether it is more pronounced in particular settings. It would also help clarify whether platform language serves as a stable anchor for partisan rhetoric or whether parties diverge more sharply in venues with fewer institutional constraints.

This study treats 2008 as a theoretically motivated starting point, given the convergence of the global financial crisis, generational change, and shifts in the international and technological environment discussed in Section 2.1. However, the analysis does not directly test whether the patterns identified here represent a structural break relative to earlier periods or an acceleration of longer-running trends in party agenda construction. Extending the corpus to include pre-2008 platforms would allow future work to evaluate whether divergence in issue clustering, internal topic structure, and emotional tone emerges discontinuously after 2008 or evolves more gradually across earlier decades. Such an extension would require additional work that falls outside the scope of the present study, but would provide a valuable robustness check on the post-2008 focus adopted here.

Finally, the analysis identifies divergence but does not directly examine its consequences. It would be very interesting to explore whether changes in platform structure predict legislative behavior, media framing or public perceptions of political issues. Linking semantic and emotional divergence to election outcomes would offer a clearer picture of how discursive shifts shape governance and political conflict. These extensions would complement the descriptive patterns identified here and help clarify the broader implications of agenda-level polarization.

6 Conclusion

This study examined how the semantic structure and rhetorical tone of U.S. party platforms evolved between 2008 and 2024. Using a multi-stage NLP pipeline, it traced changes in thematic emphasis, within-issue framing and emotional sentiment across five electoral cycles. The results show that Democrats and Republicans increasingly diverge not only in their preferred policy solutions but in the broader construction of political problems. Democrats reorganize their agenda more frequently, incorporating emerging domains such as climate change, digital rights and AI governance, while Republicans maintain a more stable rhetorical template centered on sovereignty, order and constitutional protection. Even within shared domains such as healthcare, immigration and technology, the parties structure their arguments around fundamentally different subtopics and evaluative lenses.

These findings suggest that contemporary polarization operates at the level of agenda construction as well as ideology. Rather than debating a shared set of problems, the parties increasingly describe different political realities. This shift complicates efforts at bipartisan cooperation, since disagreements begin not with preferred solutions but with incompatible definitions of what constitutes a national challenge. The analysis also highlights the value of text-as-data methods for uncovering structural patterns that are difficult to observe through conventional qualitative reading alone.

While party platforms represent only one mode of partisan communication, they provide a useful window into how organizations articulate their priorities and signal their governing commitments. Understanding how these documents evolve over time clarifies the deeper foundations of partisan conflict in the United States. By mapping semantic, structural and affective divergence across the post-2008 period, this study hope to contribute to broader

debates on polarization and offers a framework for examining how political actors reinterpret the issues that define national governance.

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Topic_ID	Count	Top_5_Words	Full_Words_with_Scores	Representative_Docs	
0	0	697	nuclear weapons, nuclear, foreign policy, sanctions, nato	nuclear weapons(0.73), nuclear(0.58), foreign policy(0.46), sanctions(0.44), nato(0.41), treaty(0.40), north korea(0.40), diplomacy(0.39), strategic(0.33), united nations(0.33)	[‘We will make the goal of eliminating nuclear weapons worldwide a central element of U.S. nuclear weapons policy.’; ‘Donald Trump encourages the spread of nuclear weapons across Asia and the Middle East, which would weaken the Treaty on the Non-Proliferation of Nuclear Weapons (NPT).’; ‘We face the growing threat of terrorists acquiring nuclear weapons or the materials to make them, as more countries seek nuclear weapons and nuclear materials remain unsecured in too many places.’]
1	1	664	human rights, voting rights, rights, freedoms, civil rights	human rights(0.80), voting rights(0.77), rights(0.73), freedoms(0.66), civil rights(0.65), democracy(0.53), right vote(0.53), laws(0.51), enforcement(0.49), accountability(0.49)	[‘and human rights’; ‘human rights’; ‘VOTING RIGHTS’]
2	2	602	health care, health insurance, affordable care, health, health outcomes	health care(0.86), health insurance(0.69), affordable care(0.67), health(0.64), health outcomes(0.64), public health(0.61), health centers(0.61), mental health(0.59), medicare(0.56), community health(0.55)	[‘For millions of Americans, that essential mental health care is out of reach.’; ‘Health Care’; ‘health care.’]
3	3	589	housing, affordable housing, public housing, fair housing, rent	housing(0.75), affordable housing(0.74), public housing(0.65), fair housing(0.61), rent(0.53), homes(0.52), infrastructure(0.49), affordable(0.47), funding(0.46), rental(0.45)	[‘, and affordable housing.’; ‘and expand housing’; ‘HOUSING’]
4	4	464	climate change, clean energy, climate crisis, energy, energy efficient	climate change(0.74), clean energy(0.73), climate crisis(0.67), energy(0.67), energy efficient(0.66), renewable energy(0.64), energy economy(0.63), renewable(0.63), combating climate(0.63), climate(0.60)	[‘Climate Change’; ‘climate change’; ‘clean energy.’]
5	5	447	discrimination, medicare, civil rights, gender identity, policies	discrimination(0.41), medicare(0.32), civil rights(0.30), gender identity(0.25), policies(0.24), orientation gender(0.23), health care(0.22), benefits(0.21), people disabilities(0.20), disabilities(0.20)	[‘At the core of the Democratic Party is the principle that no one should face discrimination on the basis of race, ethnicity, national origin, language, religion, gender, sexual orientation, gender identity, or disability status.’; ‘Democrats will fully enforce the Americans with Disabilities Act, the Individuals with Disabilities Education Act, the Fair Housing Act, the Civil Rights of Institutionalized Persons Act, Section 504 of the Rehabilitation Act, the Mental Health Parity and Addiction Equity Act, and the Help America Vote Act, among other bedrock statutes protecting the rights of people with disabilities.’; ‘Democrats are committed to ending discrimination on the basis of race, ethnicity, national origin, religion, language, gender, age, sexual orientation, gender identity, or disability status.’]
6	6	354	workers rights, union jobs, workers, discrimination, labor	workers rights(0.70), union jobs(0.55), workers(0.54), discrimination(0.52), labor(0.52), employers(0.52), collective bargaining(0.50), trade agreements(0.49), employment(0.49), employees(0.47)	[‘We will prevent employers from taking advantage of immigrant workers by establishing an affirmative process to request deferred action for workers who report labor violations and by supporting the Domestic Workers’ Bill of Rights and the Protect Our Workers from Exploitation and Retaliation (POWER) Act.’; ‘Promoting Trade That is Fair and Benefits American Workers’; ‘by American workers.’]
7	7	339	public education, public schools, charter schools, support public, funding	public education(0.47), public schools(0.47), charter schools(0.47), support public(0.44), funding(0.44), educators(0.44), education(0.42), arts(0.41), public school(0.41), humanities(0.40)	[‘We support art in schools and increased public funding for the National Endowment for the Arts and the National Endowment for the Humanities.’; ‘We will continue to support public funding for the National Endowment for the Arts, for the National Endowment for the Humanities, and for programs providing art and music education in primary and secondary schools.’; ‘Charter schools must reflect their communities, and thus must accept and retain proportionate numbers of students of color, students with disabilities and English Language Learners in relation to their neighborhood public schools.’]
8	8	315	violence women, womens rights, discrimination women, violence, womens	violence women(0.75), womens rights(0.74), discrimination women(0.55), violence(0.54), womens(0.50), women(0.50), women act(0.49), reproductive freedom(0.48), womans(0.47), equal rights(0.47)	[‘VIOLENCE AGAINST WOMEN’; ‘Protecting Women’s Rights’; ‘We’ll defend all women’s right to reproductive health care.’]
9	9	265	tax credits, tax credit, child tax, tax cuts, tax breaks	tax credits(0.64), tax credit(0.63), child tax(0.58), tax cuts(0.57), tax breaks(0.57), tax cut(0.56), tax relief(0.54), cut taxes(0.54), tax break(0.50), income tax(0.47)	[‘and expanded the Child Tax Credit and Earned Income Tax Credit.’; ‘and reduce barriers for working families to benefit from targeted tax breaks, including the Earned Income Tax Credit and the Child Tax Credit.’; ‘We will make it easier for working families to benefit from targeted tax breaks, including the Earned Income Tax Credit and the Child Tax Credit, which too often go unclaimed by the lowest-income tax filers.’]
10	10	254	puerto rico, guam, guam samoa, mariana islands, puerto	puerto rico(0.62), guam(0.61), guam samoa(0.61), mariana islands(0.57), puerto(0.56), territories(0.56), rico(0.52), people territories(0.51), virgin islands(0.50), native hawaiians(0.47)	[‘President Biden, Vice President Harris, and Democrats have made historic investments in Washington, D.C., Puerto Rico, the U.S. Virgin Islands, Guam, the Commonwealth of the Northern Mariana Islands, and American Samoa -’, ‘Puerto Rico, Guam, American Samoa, the Northern Mariana Islands, and the U.S.’; ‘Virgin Islands We recognize and honor the contributions and the sacrifices made in service of our country by the people living in Puerto Rico, Guam, American Samoa, the Northern Mariana Islands’]
11	11	227	vice harris, took office, executive order, bidens, harris	vice harris(0.50), took office(0.40), executive order(0.37), bidens(0.36), harris(0.35), white house(0.35), executive orders(0.33), leaders(0.32), renewing leadership(0.32), leadership(0.31)	[‘President Biden and Vice President Harris turned a setback into a great American comeback for working families.’; ‘Since their first days in office, President Biden and Vice President Harris have taken action to protect the sacred right to vote in free, fair, and secure elections.’; ‘President Biden and Vice President Harris have taken historic action to stand up to hate.’]
12	12	189	middle class, economy, ending poverty, economic, wealth gap	middle class(0.79), economy(0.50), ending poverty(0.47), economic(0.46), wealth gap(0.45), racial wealth(0.44), poverty(0.44), middle(0.44), living poverty(0.42), worst economic(0.42)	[‘grow the middle class.’; ‘The middle class built America -’, ‘so that they build the American middle class.’]
13	13	150	immigration reform, immigration, legal immigration, broken immigration, comprehensive immigration	immigration reform(0.76), immigration(0.73), legal immigration(0.71), broken immigration(0.68), comprehensive immigration(0.68), immigrants(0.64), undocumented immigrants(0.62), migrants(0.56), visas(0.51), deportation(0.51)	[‘We must also improve the legal immigration system’, ‘fix our immigration system.’; ‘Immigration America has always been a nation of immigrants.’]
14	14	135	families veterans, help veterans, veterans, ensure veteran, veterans service	families veterans(0.77), help veterans(0.73), veterans(0.71), ensure veteran(0.69), veterans service(0.69), veterans military(0.68), veterans affairs(0.66), department veterans(0.64), faith veterans(0.63), veteran(0.59)	[‘MEETING OUR NATION’S SACRED OBLIGATION TO VETERANS AND MILITARY FAMILIES’; ‘Supporting Troops, Military Families, and Veterans. President Obama and the Democratic Party are committed to keeping the sacred trust we have with our troops, military families, and veterans.’; ‘and benefits for our veterans and their families.’]
15	15	111	homeland security, terrorist threats, cybersecurity, cyber threats, security	homeland security(0.68), terrorist threats(0.58), cybersecurity(0.57), cyber threats(0.54), security(0.50), threats(0.49), cyber attacks(0.48), terrorism(0.47), domestic terrorism(0.47), terrorists(0.43)	[‘Homeland Security to undertake a top to bottom assessment of the threats we face and our’, ‘Cybersecurity, Cybersecurity threats represent one of the most serious potential national security, public safety, and economic challenges we face.’; ‘Pursue Intelligence Reform To succeed, our homeland security and counter-terrorism actions must be linked to an intelligence community that deals effectively with the threats we face.’]

Figure 12: Democrat full topic distribution

16	16	68	drug prices, drug costs, prescription drugs, prescription drug, drug companies	prescription drugs(0.63), prescription drug(0.62), drug companies(0.58), pharmaceutical companies(0.57), prescription(0.55), pharmaceutical(0.54), drugmakers(0.51), lower drug(0.49)	[reduces prescription drug prices, premiums, and out-of-pocket costs', 'reduce prescription drug prices by standing up to big pharmaceutical companies', 'and we stood up to Big Pharma to lower prescription drug prices.]
17	17	62	religious freedom, freedom religion, protect religious, religious minorities, religious	religious freedom(0.82), freedom religion(0.78), protect religious(0.74), religious minorities(0.73), religious(0.58), religion(0.56), protecting freedom(0.53), freedom(0.53), muslims(0.51), right freedom(0.51)	[We will advocate for religious freedom across the world.', 'Religious Minorities', 'religious freedom,]
18	18	59	privacy rights, privacy, data privacy, protect consumers, privacy civil	privacy rights(0.75), privacy(0.73), data privacy(0.66), protect consumers(0.61), privacy civil(0.60), protect individuals(0.57), protecting(0.54), protect(0.53), protections(0.52), protect americas(0.50)	[Protecting Consumer Rights and Privacy', 'protect the privacy of Americans', 'protect Americans' privacy,]
19	19	59	child care, affordable child, childcare, working families, millions children	child care(0.65), affordable child(0.58), childcare(0.57), working families(0.57), millions children(0.55), supporting families(0.54), healthy families(0.51), care providers(0.48), parents(0.48), families(0.47)	[and 220,000 child care centers serving millions of kids, which working parents rely on.', 'And because affordable child care is critical to expanding opportunities for parents, especially women, President Biden helped hundreds of thousands of child care providers keep their doors open to millions of children', 'and to support affordable child care, so parents are not forced to choose between keeping their jobs and keeping their children safe.]
20	20	54	wall street, street, main street, rules wall, reform	wall street(0.75), street(0.52), main street(0.46), rules wall(0.40), reform(0.40), antitrust laws(0.39), economic(0.38), wall(0.37), antitrust(0.36), economy(0.34)	[Curbing Wall Street Abuses', 'and when the same rules apply to everyone, from Main Street to Wall Street.', 'Wall Street Reform]
21	21	37	appoint judges, appointing judges, judicial appointees, court justices, justices	appoint judges(0.81), appointing judges(0.78), judicial appointees(0.75), court justices(0.73), justices(0.72), judges(0.72), appointed(0.60), appointees(0.59), supreme court(0.58), judicial(0.57)	[Judges should reflect the diverse population whose laws they shape, and President Biden has appointed more than 200 highly qualified federal judges who look like America.', 'JUDGES', 'We will appoint U.S. Supreme Court justices and federal judges who look like America.]
22	22	30	trump refuses, political violence, place trump, quit fighting, bush administration	trump refuses(0.47), political violence(0.38), place trump(0.32), quit fighting(0.30), bush administration(0.27), democracy(0.26), rhetoric(0.26), election(0.25), political(0.25), bush(0.25)	[President Biden, Vice President Harris, and Democrats have consistently made it clear that political violence has no place in America.', 'and making it clear that political violence has no place in America.', 'He still doesn't accept the plain truth, upheld by scores of courts, that he lost in 2020, even describing himself as a "very proud election denier" -- and he has never agreed to accept this year's results.]
23	23	29	ai safety, ai, standards ai, artificial intelligence, intelligence	ai safety(0.73), ai(0.69), standards ai(0.62), artificial intelligence(0.51), intelligence(0.45), artificial(0.38), institute(0.36), funding critical(0.34), institutes(0.34), technologies(0.33)	[and created the AI Safety Institute at the National Institute of Standards and Technology (NIST) to set rigorous standards for extensive testing of powerful AI models to ensure safety before public release.', 'and mitigating AI risk', 'We will also prioritize funding for critical AI research and development that advances AI safety.]
24	24	23	middle east, europe middle, africa, north africa, syria	middle east(0.91), europe middle(0.69), africa(0.67), north africa(0.66), syria(0.65), africa africa(0.64), western hemisphere(0.57), europe(0.57), russia(0.51), china(0.51)	[Middle East', 'THE MIDDLE EAST & NORTH AFRICA', 'Europe 79 Indo-Pacific 81 China 82 The Middle East & North Africa 84 Western Hemisphere 87 Africa 88 Strengthening]
25	25	23	faith community, people faiths, city accessible, faithbased groups, places worship	faith community(0.57), people faiths(0.56), city accessible(0.55), faithbased groups(0.53), places worship(0.53), faith country(0.52), faithbased(0.50), faiths(0.49), faith(0.49), undivided city(0.48)	[an undivided city accessible to people of all faiths.', 'It should remain an undivided city accessible to people of all faiths.', 'It should remain an undivided city accessible to people of all faiths.]
26	26	22	postal service, postal services, usps, postal, including postal	postal service(0.77), postal services(0.76), usps(0.69), postal(0.68), including postal(0.58), mail(0.50), services strengthening(0.38), strong public(0.37), delivery(0.36), metropolitan(0.30)	[The U.S. Postal Service (USPS) is the world's most efficient mail carrier.', 'Strengthening the U.S. Postal Service', 'The United States Postal Service (USPS) is a national treasure.]
27	27	21	repeal affordable, promising repeal, repeal, affordable care, pandemic trump	repeal affordable(0.65), promising repeal(0.57), repeal(0.54), affordable care(0.50), pandemic trump(0.45), trump wants(0.44), health care(0.42), trump tried(0.40), care act(0.39), trump(0.37)	[He will repeal the Affordable Care Act, putting coverage of prenatal, pregnancy, and postpartum care at risk', 'Trump wants to repeal the Affordable Care Act, which will increase costs and rip away protections for LGBTQ+ Americans', 'Now, he wants to repeal the Affordable Care Act, ripping health care away from millions.]
28	28	20	registered apprenticeships, apprenticeship programs, apprenticeships, apprenticeship, training workers	registered apprenticeships(0.87), apprenticeship programs(0.82), apprenticeships(0.80), apprenticeship(0.76), training workers(0.62), job training(0.60), training opportunities(0.59), training programs(0.57), training(0.54), training administration(0.50)	[We have a million trainees enrolled in registered apprenticeships today.', 'and address the changing climate, including through pre-apprenticeship opportunities, joint labor-management registered apprenticeships for training, and direct-hire programs', 'as well as in registered apprenticeships,]
29	29	15	fentanyl, drugs, drugs like, cannabis, traffickers	fentanyl(0.72), drugs(0.36), drugs like(0.33), cannabis(0.31), traffickers(0.30), trafficking illicit(0.27), mexico(0.26), illicit(0.26), border(0.24), trafficking(0.23)	[President Biden has made fighting the trafficking of fentanyl a top priority.', 'We're seizing record amounts of fentanyl', 'He has seized record amounts of fentanyl at our border—]
30	30	14	cuba, people cuba, venezuela, build obamas, alliances	cuba(0.62), people cuba(0.60), venezuela(0.33), build obamas(0.28), alliances(0.26), universal rights(0.25), greater freedom(0.25), obamas(0.25), people support(0.24), remittances(0.24)	[We have taken additional steps to bolster Cuban civil society, expanding purposeful exchanges that bolster independent religious groups on the island and enhancing the free flow of information to, from, and among the Cuban people.', 'Because of steps the President has taken, it is now possible for Cuban Americans to visit and support their families in Cuba, and to send remittances that reduce the Cuban people's dependence on the Cuban state.', 'And we must build ties to the people of Cuba and help advance their liberty by allowing unlimited family visits and remittances to the island']
31	31	13	scientists, scientific, scientific research, science, based science	scientists(0.60), scientific(0.55), scientific research(0.53), science(0.53), based science(0.52), research agencies(0.46), efforts undermine(0.46), health science(0.45), scientific discovery(0.45), trump administrations(0.44)	[the rejection of science.', 'The Trump Administration's systematic efforts to undermine, discredit, dismiss, fire, disempower, and diminish the contributions of scientists, in health science and other disciplines, have weakened our public institutions.', 'Democrats oppose the Trump Administration's efforts to undermine federal scientific research institutions, apply political screens to research grants, and interfere with the work of career scientists—actions which have a chilling effect on scientific discovery]
32	32	12	safety standards, workplace safety, health safety, safety health, worker safety	safety standards(0.72), workplace safety(0.68), health safety(0.67), safety health(0.65), worker safety(0.63), standards ensuring(0.62), appropriate safety(0.58), safety rules(0.57), standards protect(0.57), safety(0.53)	[including by issuing and enforcing effective infectious disease workplace safety standards through the Occupational Safety and Health Administration.', 'Our Occupational Safety and Health Administration will continue to adopt and enforce comprehensive safety standards.', 'Our Occupational Safety and Health Administration will adopt and enforce comprehensive safety standards.]
33	33	12	finish job, work job, finish, far	finish job(0.58), work(0.55), job(0.46), finish(0.44), far(0.31), time(0.27), stop(0.27), (0.23), (0.23), (0.23)	[There is more work to do.', 'We have to finish the job.', 'Now, it's time to finish the job.]

Figure 13: Democrat full topic distribution

Topic_ID	Count	Top_5_Words	Full_Words_with_Scores	Representative_Docs	
0	0	734	domestic energy, renewable energy, independence, energy, fuels	domestic energy(0.66), renewable(0.62), energy independence(0.62), energy(0.59), fuels(0.56), coal(0.47), fuel(0.45), natural gas(0.42), economy(0.42), economic(0.42)	['Although alternate fuels will shape our energy future, coal — America's most affordable and abundant energy resource and the source of most of our electricity — remains a strategic national resource that must play a major role in energy independence.', 'energy development', 'Our Nation's Energy Abundance']
1	1	654	fifth amendment, amendment right, inalienable rights, amendment, liberties	fifth amendment(0.78), amendment right(0.71), inalienable rights(0.71), amendment(0.64), liberties(0.61), rights(0.61), constitution(0.59), property rights(0.57), constitutional(0.54), constitutions(0.53)	['The Fifth Amendment: Intellectual Property Rights', 'The Fifth Amendment: Intellectual Property Rights', 'a right which antedated the Constitution and was solemnly confirmed by the Second Amendment.']
2	2	403	immigration, illegal immigration, borders, illegal aliens, border	immigration(0.65), illegal immigration(0.58), borders(0.49), illegal aliens(0.46), border(0.44), southern border(0.42), security interests(0.41), illegal alien(0.41), visas(0.41), sovereignty(0.40)	['While immigration is addressed in more detail elsewhere, we cannot ignore the reality that border security is a national security issue.', 'While immigration is addressed in more detail elsewhere, we cannot ignore the reality that border security is a national security issue.', 'Immigration, National Security, and the Rule of Law']
3	3	379	voting rights, voter, elections, voting, ballot	voting rights(0.66), voter(0.65), elections(0.63), voting(0.60), ballot(0.59), voters(0.58), election(0.57), vote(0.53), integrity(0.45), voted(0.37)	['Honest Elections and the Right to Vote', 'We support efforts by state and local election officials to ensure integrity in the voting process and to prevent voter fraud and abuse, particularly as it relates to voter registration and absentee ballots.', 'Voter Integrity to Ensure Honest Elections']
4	4	378	health care, health insurance, patients, care providers, healthcare	health care(0.62), health insurance(0.60), patients(0.58), care providers(0.57), healthcare(0.55), affordable care(0.50), obamacare(0.47), medicare(0.47), medicaid(0.46), physicians(0.45)	['Patients deserve access to health care providers they trust who will personalize and coordinate their care to ensure they receive the right treatment with the right health care provider at the right time.', 'Patients must have information to make sound decisions about their health care providers, hospitals, and insurance companies.', 'Health Care Reform: Putting Patients First']
5	5	372	trade agreements, world trade, global economy, treaty, treaties	trade agreements(0.74), world trade(0.67), global economy(0.56), treaty(0.53), treaties(0.49), economy(0.47), multilateral(0.47), agreements(0.46), trading(0.44), economic(0.44)	['The territories' economic stability and potential for growth must be considered in any trade agreements between the United States and other Pacific nations.', 'The territories' economic stability and potential for growth must be considered in any trade agreements between the United States and other Pacific nations.', 'Rebalance Trade']
6	6	343	restructuring tax, taxation, tax, tax rates, taxes	restructuring tax(0.65), taxation(0.62), tax(0.60), tax rates(0.60), taxes(0.59), tax cuts(0.59), tax relief(0.55), tax code(0.52), isrs(0.51), taxpayer(0.49)	['Today's tax code is tilted against them, with one of the highest corporate tax rates of all developed countries.', 'In any restructuring of federal taxation, to guard against hypertaxation of the American people, any value added tax or national sales tax must be tied to the simultaneous repeal of the Sixteenth Amendment, which established the federal income tax.', 'AND NO TAX ON TIPS!']
7	7	286	farmers ranchers, farmers, ranchers, agriculture, agricultural	farmers ranchers(0.73), farmers(0.66), ranchers(0.61), agriculture(0.60), agricultural(0.59), ranching(0.58), department agriculture(0.49), agricultural exports(0.48), public lands(0.47), farming(0.46)	['Farmers and ranchers are among this country's leading conservationists.', 'This includes providing for an abundant water supply for America's farmers, ranchers, and their communities.', 'In the interest of protecting the safety of our farmers and ranchers, we will advance policies to protect the security, privacy, and most of all, the private ownership of individual farmers' and ranchers' data.'].']
8	8	267	israel, middle east, hezbollah, isis, iran	israel(0.70), middle east(0.66), hezbollah(0.51), isis(0.50), iran(0.49), iraq(0.46), sanctions(0.45), diplomacy(0.43), nuclear weapons(0.43), nuclear arsenal(0.43)	['Israel is a vigorous democracy, unique in the Middle East.', 'The Middle East', 'and seek peace in the Middle East.'].']
9	9	238	veterans, returning veterans, vets, va, troops	veterans(0.76), returning veterans(0.71), vets(0.51), va(0.51), troops(0.43), soldiers(0.42), military(0.40), military families(0.39), care(0.39), medical care(0.38)	['The work of the Department of Veterans Affairs (VA) is essential to meet our commitments to them: Providing health, education, disability, survivor, and home loan benefits, and arranging memorial services upon death.', 'cares for Veterans.', 'Take Care of Our Veterans']
10	10	215	honor, patriotism, immigrants, gratitude, immigrant	honor(0.45), patriotism(0.43), immigrants(0.41), gratitude(0.40), immigrant(0.39), citizenship(0.39), indians(0.38), legal immigrants(0.37), military service(0.36), citizens(0.36)	['We honor that commitment, loyalty, and sacrifice of all American Indians serving in the military today and in years past', 'We are particularly grateful to the thousands of new legal immigrants, many of them not yet citizens, who are serving in the Armed Forces and among first responders.', 'We are particularly grateful to the thousands of new legal immigrants, many of them not yet citizens, who are serving in the Armed Forces and among first responders.'].']
11	11	214	higher education, education, education programs, schooling, secondary education	higher education(0.83), education(0.76), education programs(0.69), schooling(0.66), secondary education(0.64), access education(0.64), education support(0.60), worldclass education(0.57), students(0.57), academic achievement(0.57)	['Choice in Education', 'Choice in Education', 'Higher Education']
12	12	158	abortion, oppose funding, oppose, affirm unborn, fetal	abortion(0.62), oppose funding(0.50), oppose(0.42), affirm unborn(0.41), fetal(0.38), pregnancy(0.38), unborn(0.38), unborn child(0.37), opposition(0.34), providers(0.32)	['Because of their opposition to simple abortion clinic safety procedures, support for taxpayer-funded abortion, and rejection of pregnancy resource centers that provide abortion alternatives.', 'Because of their opposition to simple abortion clinic safety procedures, support for taxpayer-funded abortion, and rejection of pregnancy resource centers that provide abortion alternatives.', 'We call for a permanent ban on federal funding and subsidies for abortion and healthcare plans that include abortion coverage.'].']
13	13	156	religious liberty, religious freedom, rights religious, rights conscience, freedom	religious liberty(0.78), religious freedom(0.71), rights religious(0.71), rights conscience(0.55), freedom(0.53), liberty(0.51), amendment(0.46), religious moral(0.45), rights(0.44), religious persecution(0.42)	['The First Amendment: Religious Liberty', 'The Bill of Rights lists religious liberty, with its rights of conscience, as the first freedom to be protected.', 'The Bill of Rights lists religious liberty, with its rights of conscience, as the first freedom to be protected.'].']
14	14	131	current administrations, administrations, current administration, foreign policy, administration	current administrations(0.56), administrations(0.49), current administration(0.46), foreign policy(0.38), administration(0.38), white house(0.37), leadership(0.37), americas interests(0.36), obama(0.34), bureaucratic state(0.32)	['The current Administration has abandoned America's friends and rewarded its enemies.', 'Thanks in large part to a lack of leadership from the current Administration and congressional Democrats.', 'Thanks in large part to a lack of leadership from the current Administration and congressional Democrats.'].']
15	15	122	banks, lenders, lending, loans, mortgages	banks(0.59), lenders(0.56), lending(0.56), loans(0.55), mortgages(0.53), fannie mae(0.52), bank(0.50), financial institutions(0.47), borrowers(0.43), borrowing(0.42)	['Still, the majority of agricultural loans and small business loans are made by community banks.', 'We will end the government mandates that required Fannie Mae, Freddie Mac, and federally- insured banks to satisfy lending quotas to specific groups.', 'We will end the government mandates that required Fannie Mae, Freddie Mac, and federally- insured banks to satisfy lending quotas to specific groups.'].']
16	16	113	internet safe, control internet, internet access, internet internet, privacy	internet safe(0.74), control internet(0.65), internet access(0.57), internet internet(0.56), privacy(0.53), internet(0.50), security(0.50), fundamental freedoms(0.49), protecting(0.49), freedom(0.47)	['Protecting Internet Freedom', 'Protecting Internet Freedom', 'Protecting Internet Freedom']
17	17	99	raise children, strong families, families, burdens parenting, adoption	raise children(0.63), strong families(0.62), families(0.61), burdens parenting(0.61), adoption(0.58), families communities(0.57), millions families(0.55), parenting(0.54), care children(0.51), vulnerable children(0.51)	['and recognize the burdens on families with children.', 'Deprived of a parent through no fault of their own, youngsters from these families should be a special concern of our schools, social services, and religious institutions.', 'Deprived of a parent through no fault of their own, youngsters from these families should be a special concern of our schools, social services, and religious institutions.'].']

Figure 14: Republican full topic distribution

18	18	80	oppose discrimination, discrimination, protected discrimination, fairness doctrine, nondiscrimination laws	oppose discrimination(0.71), discrimination(0.60), protected discrimination(0.55), fairness doctrine(0.47), nondiscrimination laws(0.46), enforcing nondiscrimination(0.42), unions(0.42), discriminating(0.39), enforcing(0.38), so-called fairness(0.37)	[We consider discrimination based on sex, race, age, religion, creed, disability, or national origin to be immoral, and we will strongly enforce anti-discrimination statutes.]; Therefore, we oppose discrimination based on race, sex, religion, creed, disability, or national origin and support statutes to end such discrimination.]; Therefore, we oppose discrimination based on race, sex, religion, creed, disability, or national origin and support statutes to end such discrimination.];
19	19	71	housing market, housing, housing administration, recession, households	housing market(0.73), housing(0.55), housing administration(0.51), recession(0.48), households(0.43), mortgage(0.42), homes(0.39), economy(0.35), homeownership(0.35), economic viability(0.33)	[We must scale back the federal role in the housing market.]; 'We must scale back the federal role in the housing market.]; The collapse of the housing market over the last four years has been not only a severe blow to the entire economy, but also a personal tragedy to millions of Americans whose homes have lost value and to so many others who have lost their homes.];
20	20	70	social security, retirement, pension, retirees, retire	social security(0.91), retirement(0.52), pension(0.49), retirees(0.43), retire(0.42), nearing retirement(0.41), retirees nearing(0.39), preserve social(0.39), security(0.38), saving(0.36)	['Saving Social Security', 'Protect Social Security', 'Social Security']
21	21	69	commitment, dedication, proposal, mission, maintaining	commitment(0.86), dedication(0.51), proposal(0.47), mission(0.41), maintaining(0.41), progress(0.41), acknowledge(0.39), continue(0.38), stay(0.36), missions(0.36)	['Our Commitment', 'Our Commitment', 'Our Commitment:']
22	22	66	traditional marriage, marriage family, marriage, traditional, couples	traditional marriage(0.87), marriage family(0.67), marriage(0.63), traditional(0.51), couples(0.49), married(0.42), family(0.35), family life(0.35), household(0.34), cultural values(0.33)	['Preserving Traditional Marriage', 'Traditional marriage and family, based on marriage between one man and one woman, is the foundation for a free society and has for millennia been entrusted with rearing children and instilling cultural values.']; 'Traditional marriage and family, based on marriage between one man and one woman, is the foundation for a free society and has for millennia been entrusted with rearing children and instilling cultural values.];
23	23	57	tribal governments, tribal communities, country tribal, tribes, tribal	tribal governments(0.77), tribal communities(0.58), country tribal(0.57), tribes(0.53), tribal(0.50), indians(0.49), tribe(0.49), encourage indian(0.48), alaska natives(0.47), natives(0.46)	['Based on both treaty and other law, the federal government has a unique government-to- government relationship with and trust responsibility for Indian Tribal Governments, American Indians, and Alaska Natives.']; 'American Indians have established elected tribal governments to carry out the public policies of the tribe, administer services to its tribal member constituents, and manage relations with federal, state, and local governments.']; 'American Indians have established elected tribal governments to carry out the public policies of the tribe, administer services to its tribal member constituents, and manage relations with federal, state, and local governments.];
24	24	56	platform, reaffirm principles, goals, sections, goal	platform(0.46), reaffirm principles(0.35), goals(0.35), sections(0.34), goal(0.33), common purpose(0.33), purpose(0.31), determination(0.30), based principles(0.30), principles(0.29)	['That is why the many sections of this platform affirm our trust in the people, our faith in their judgment, and our determination to help them take back their country.']; 'That overarching goal unites all the sections of this platform.']; 'That overarching goal unites all the sections of this platform.];
25	25	55	taiwan china, taiwan, principle, principles, china	taiwan china(0.54), taiwan(0.51), principle(0.33), principles(0.33), china(0.31), chinas(0.31), chinese people(0.29), allies(0.28), oppose(0.28), human rights(0.27)	['If China were to violate those principles, the United States, in accord with the Taiwan Relations Act, will help Taiwan defend itself.']; 'If China were to violate these principles, the U.S., in accord with the Taiwan Relations Act, will help Taiwan defend itself.']; 'If China were to violate those principles, the U.S., in accord with the Taiwan Relations Act, will help Taiwan defend itself.];
26	26	54	including addiction, mental health, addiction mental, nonviolent offenders, mental illness	including addiction(0.49), mental health(0.46), addiction mental(0.46), nonviolent offenders(0.44), mental illness(0.43), drugs(0.40), opioid(0.38), war drugs(0.37), drug(0.36), prescription(0.36)	['This is especially the case with mental health care, which often amounts to prescribing drugs because there are not enough psychologists and psychiatrists to do anything else.']; 'Modifications to it should be targeted toward particular categories, especially nonviolent offenders and persons with drug, alcohol, or mental health issues.']; 'Modifications to it should be targeted toward particular categories, especially nonviolent offenders and persons with drug, alcohol, or mental health issues.];
27	27	40	opportunity act, disabilities, disabilities education, individuals disabilities, students disabilities	opportunity act(0.55), disabilities(0.50), disabilities education(0.48), individuals disabilities(0.48), students disabilities(0.46), persons disabilities(0.43), people disabilities(0.43), education act(0.42), opportunity society(0.40), opportunities(0.40)	['In addition, our Workforce Innovation and Opportunity Act will make it easier for students with disabilities to pursue competitive employment.']; 'Their Workplace Innovation and Opportunity Act — modernizing workforce programs, repealing mandates, and advancing employment for persons with disabilities — is now law.']; 'Their Workplace Innovation and Opportunity Act — modernizing workforce programs, repealing mandates, and advancing employment for persons with disabilities — is now law.];
28	28	38	puerto rico, statehood, puerto, rico, united citizens	puerto rico(0.57), statehood(0.54), puerto(0.53), rico(0.49), united citizens(0.42), citizens(0.35), caribbean(0.35), territory(0.34), support right(0.33), ratified(0.33)	['We support the right of the United States citizens of Puerto Rico to be admitted to the Union as a fully sovereign state after they freely so determine.']; 'We support the right of the United States citizens of Puerto Rico to be admitted to the Union as a fully sovereign state.']; 'We support the right of the United States citizens of Puerto Rico to be admitted to the Union as a fully sovereign state.];
29	29	31	poverty, poor, increase healthcare, labor, labor force	poverty(0.54), poor(0.36), increase healthcare(0.35), labor(0.35), labor force(0.33), population(0.31), aging population(0.31), participation(0.31), charitable(0.30), 50 years(0.30)	['We have been fighting the War on Poverty for 50 years and poverty is winning.']; 'Our labor force participation rate has plunged to at or below 63 percent for far too long, lows last seen two generations ago before President Reagan was elected in 1980.']; 'Our labor force participation rate has plunged to at or below 63 percent for far too long, lows last seen two generations ago before President Reagan was elected in 1980.];
30	30	24	nasa, space, publicprivate partnerships, partnerships, missions	nasa(0.58), space(0.50), publicprivate partnerships(0.43), partnerships(0.39), missions(0.38), mission(0.36), americas future(0.36), americas leadership(0.35), partnership(0.34), technological progress(0.33)	['The public-private partnerships between NASA, the Department of Defense, and commercial companies have given us technological progress that has reduced the cost of accessing space, "and extended America's space leadership in the commercial, civil, and national security spheres.", "and extended America's space leadership in the commercial, civil, and national security spheres.];
31	31	16	refugees, refugee, refuge, dictator, persecution	refugees(0.61), refugee(0.57), refuge(0.34), dictator(0.27), persecution(0.26), admitted country(0.25), hundreds thousands(0.20), religious persecution(0.20), thousands(0.18), hundreds(0.17)	['To ensure our national security, refugees who cannot be carefully vetted cannot be admitted to the country.']; 'The dictator of Syria, Bashar Assad, has murdered hundreds of thousands of his own people and created millions of refugees.']; 'The dictator of Syria, Bashar Assad, has murdered hundreds of thousands of his own people and created millions of refugees.];
32	32	15	taiwan, russia, asia, taiwan china, china	taiwan(0.89), russia(0.86), asia(0.85), taiwan china(0.85), china(0.84), india(0.71), vietnam(0.68), iran(0.67), pakistan(0.67), israel(0.65)	['Taiwan', 'Russia', 'Russia']
33	33	12	missile defense, missile defenses, deploy missile, russia defense, missile	missile defense(0.77), missile defenses(0.76), deploy missile(0.67), russia defense(0.63), missile(0.62), defense installations(0.55), multilayered missile(0.54), defense neglected(0.53), planned interceptors(0.48), defenses protect(0.46)	['We must fund, develop, and deploy a multi-layered missile defense system.']; 'building an Iron Dome Missile Defense Shield.']; 'In tandem with this, the current Administration has systematically undermined America's missile defense, abandoning the missile defense bases in Poland and the Czech Republic, reducing the number of planned interceptors in Alaska, and cutting the budget for missile defense.];

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